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ASSESSMENT OF BIODIVERSITY AND GROWTH PARAMETERS OF MIYAWAKI FOREST OF SELECTED SITES IN THIRUVANANTHAPURAM DISTRICT OF KERALA

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ABSTRACT

Miyawaki afforestation is a reliable forest restoration method using native plants, proposed by Professor Akira Miyawaki in the 1980s, and introduced in Japan with the challenge to restore indigenous ecosystems and maintain global environments, which helps in disaster prevention and carbon dioxide (CO₂) mitigation. The Miyawaki micro-afforestation aims to grow many trees in a limited space in a brief period of time and grow a forest similar to a natural forest. Today, Miyawaki method has been used successfully to restore the forest worldwide. A study was conducted to find the growth performance of selected plant species grown as per the Miyawaki method of planting in three sites of the Thiruvananthapuram district of Kerala. The study site includes Miyawaki forest in Kanakakunnu (site 1), Government High School Chala (site 2), and Dept. of Botany, Kariavattom (site 3). The growth parameters of the selected plants in the sites were recorded. The growth rate was compared to the naturally growing plant species, which indicated that the plants in the Miyawaki forest in three sites had a higher growth rate. These results could offer a chance to introduce the advantages of a new afforestation method, which helps to reduce the time for a complete environmental restoration in a short period. Moreover, it also offers a model for conserving native plant species within their natural habitat, even in urban areas. It can also be concluded that these native micro forests can offer many ecological and social services, including protecting many pollinators and conserving biodiversity in urban areas. Thus the use of these indigenous species can be successfully employed for the Miyawaki afforestation technique to create thick, multilayered forests.

Keywords: Restoration, forest, Miyawaki, growth performance

INTRODUCTION

Forest is the most stable type of natural ecosystem, given suitable moisture, temperature, and soil condition. They can provide each and every necessity in a sustainable manner (Schuldt & Scherer-Lorezen, 2014). The rapid extent of exploitation in its cover due to colonization, urbanization and agriculture has affected the ecological balance, threatening the survival of plants and animals (Elliott & Swank, 1994; Mohd-Azlan et al. 2018). The creation of forests is a fundamental ecological method for restoring the natural environment, especially in urban areas, which are dominated by non-biological materials. People are taking action at local, national, and regional levels to save forest biodiversity around the globe (McAlpine and Ahmed Djoghla, 2010). Today, trees in urban areas are being planted for many different reasons. Aesthetic beautification is one main purpose in urban environments, especially those around industrial sites and transportation corridors. Taking several hundred years to complete the process of forest restoration is too long for us; because we live in a world where industry and urbanization are developing very rapidly, the improvement of an alternative reforestation technique that reduces these times could be a useful tool (Miyawaki 1993).

One reliable forest restoration method is the “native forests by native trees,” based on the vegetation–ecological theories (Miyawaki 1993; Miyawaki and Golley 1993; Padilla and Pugnaire 2006) proposed by Prof. Akira Miyawaki and applied first in Japan. According to this method, restoring native green environments, multilayer forests, and natural biocoenosis is possible, and well-developed ecosystems can be quickly established because of the simultaneous use of intermediate and late successional species in plantations (Schirone et al., 2011). The Miyawaki method involves surveying the potential natural vegetation of the area to be reforested and recovering topsoil to a depth of 20– 30 cm by mixing the soil and compost from organic materials, such as fallen leaves, mowed grass, etc. In this way, the time of the natural process of soil evolution, established by the vegetational succession itself, is reduced. The micro forests create their own ecosystem comprising several birds, butterflies, and insects. The Miyawaki micro-afforestation project is a solution for the mitigation of pollution and rise in temperature in the urban area, and it also provides many ecological and social services, such as disaster prevention and mitigation.

Deforestation and the destruction of ecosystems in the label of development lead to natural disasters and climate change. For the last four years, Kerala has been witnessing a series of natural disasters. It is imperative for the state now to embrace environment restoration methods to avoid another disaster and to rebuild the lost biodiversity. The Miyawaki model can be successfully implemented in the landlocked state of Kerala. The first Miyawaki project in Kerala was launched on January 31st, 2018, on the private land at Puliarakonam, Thiruvananthapuram. In this site, various vegetation types have already grown and turned into a forest. Implementing this project in urban areas of Kerala will increase the tree covers for the restoration of degraded forests. Only a little works were carried out in Kerala on different aspects of the Miyawaki forest. The present investigation was undertaken to assess the biodiversity and find the growth performance of selected plant species in three Miyawaki sites in Thiruvananthapuram district of Kerala, Kanakakunnu, Government High School Chala, and Dept. of Botany, University of Kerala, Kariavattom.

METHODOLOGY

Study sites

The first plot marked for the study was the Kanakakunnu palace Miyawaki forest, which was planted on 2nd January, 2019 (**Latitude:** 8°30'32.22"N, **Longitude:** 76°57'27.40"E, **Altitude:** 21 m) (Fig.1 A). The forest covers up to 5 cents (21 m × 7.41 m), containing about 426 plants belonging to 120 genera. Specially prepared organic fertilizers were used for plant growth, and they were applied only once to these plants. Insecticides were utterly avoided, and the plants grew naturally. This model micro forest was developed for the Kerala travel and tourism department by Nature's Green Guardian Foundation, Invis Multimedia Pvt Ltd., Cultural Shoppy Pvt. Ltd and Organic Kerala Mission society as a part of their social responsibility program.

The second plot selected for the study was Miyawaki forest in Government High School, Chala, Thiruvananthapuram (**Latitude:** 8°28'52.71"N, **Longitude:** 76°57'28.81"E, **Altitude:** 20 m) (Fig.1 B). The samplings were planted on 29th January 2020. According to the Miyawaki system, plants belonging to particular regions were selected and grown; apart from that, about 10% of exotic trees were also grown for insects and birds. The soil was initially made fertile by applying organic manure without the use of chemical fertilizer and

pesticides. Trees were also planted, and they were grown naturally. These 10 cents (26.8m × 13.20m) of land contains almost 1603 plants grown as per the Miyawaki system. Collaborating with the Kerala Development and Innovation Strategic Council, Culture Shoppe, and the Nature Green Guardian Foundation, launched the Miyawaki Project at Government High School, Chala, Thiruvananthapuram, on Dr.Akira Miyawaki's 92nd birthday.

The third study site included the micro forest Miyawaki garden; Department of Botany, University of Kerala, Kariavattom initiated on June 21st 2021 (**Latitude:** 8°33'59.59"N, **Longitude:** 76°53'11.11"E, **Altitude:** 22 m). It covers an area of 14.60 m × 14.50 m, and the field was prepared by proper leveling, and seedlings were planted at proper distances. About 800 plants were grown as per the Miyawaki method. All other agricultural practices such as irrigation and fertilization were uniformly applied to the forest.

Data collection

The method included planting trees with only native species in the same area, and the saplings became self-sustainable. The plants grown in the selected area were identified and authenticated with the help of flora and taxonomic experts. The categorization of plants was done based on their habits like shrubs, herbs, trees, and canopy for further documentation. Different growth parameters like plant height and girth, number of leaves, and number of branches were studied through regular field visits.

Data Analysis

Different aspects of quantitative growth parameters were analyzed statistically. The statistical data analyses were performed using IBM SPSS Statistics 22 to elucidate growth parameters between the plant species. To assess the significance of quantitative morphological characters, one-way ANOVA was done, and the results were expressed as mean ± standard error.

RESULTS AND DISCUSSION

About thirty-seven plant species were selected for the study from site 1 Kanakakunnu and were listed in table 1; it includes Trees, Shrubs, and Herbs. The plants were categorized into habitat and growth patterns, and advantages. The growth parameters such as the height of the

plant, girth of the plant, number of leaves, and number of primary branches were observed continuously at an interval of one week. The data were recorded for up to five weeks. The growth rate/height of the trees, herbs, and shrubs was plotted in figure 1-3. The girth of the plants showed significant variations in trees, shrubs, and herbs in figure 4-6. All the plants showed a higher growth rate with more leaves and branches. The number of leaves and the number of primary branches were counted, and the data were analyzed (Figure 7-12). *Lannea coromandelica* showed the highest growth rate ranging from 10.17-10.59m. Similarly, growth of the grass species is higher in *Ochlandra* sp. (4.18-4.32m) among shrubs. Among herbs, *Commelina* sp. has the highest growth rate ranging from 2.25-2.32 m.

In the second field, Miyawaki forest in Government High School Chala, Thiruvananthapuram, about fifty-eight plant species were selected for the study and listed in table 2. The plants were categorized as mentioned above, and growth parameters were observed continuously at an interval of one week. The data were recorded for up to five weeks, and the result obtained shows that the growth of the plants was faster than the usual field plant. The growth rate/height, girth, number of primary branches, and number of leaves of the trees, shrubs, herbs, and climbers were plotted in figure 13-28. The number of leaves becomes more, and the leaf with wide lamina helps to increase the biomass of the plant productivity. Thick forest and leaf prevent over sunlight into the field; it helps the field becomes wet. The micro forest is a shelter for many insects and birds; insects help in the pollination process. The plant visitors or pollinators were observed frequently and were listed in Table 4 (Plate. 2). The fertile soil is enriched with fungal diversity; fungi were collected and identified with taxonomic experts and were listed in Table 5 (Plate. 3). *Sesbania grandiflora* possesses the highest growth rate (6.28-6.44 m) among tree species. Among shrubs, *Calotropis gigantea* (4.52-4.78 m), and among herbs, *Plumbago zeylanica* (1.9-2.03 m), showed the highest growth rate in plot 2.

The third study site included the micro forest Miyawaki garden; Department of Botany, University of Kerala, Kariavattom. About thirty-eight plant species were selected for the study and listed in table 3; it includes trees, shrubs, and herbs. The plants were categorized as mentioned above, and growth parameters such as the height of the plant, girth of the plant, number of leaves, and numbers of primary branches were observed continuously every week. The data were recorded for up to five weeks, and the result obtained shows that the growth of

the plants was faster than the usual field plant. The growth rate/height, girth, number of primary branches, and number of leaves of the trees, shrubs, and herbs were plotted in Figures 29-40. *Annona reticulata* (1.412-1.418) was the highest growing plant when compared to other trees species on the site. Among shrubs, *Lawsonia inermis* (0.902-1.159 m) showed the highest growth, while among herbs, *Coleus amboinicus* (0.262-0.481 m) recorded the highest growth rate.

The growth rate of naturally growing trees species *Artocarpus heterophyllus*, *Syzygium cumini*, *Phyllanthus* sp., *Mangifera indica*, *Morus alba*, *Ixora* sp. were recorded after one year and were tabulated (Table .4). The growth rate was compared to the naturally growing plant species, and results indicated that the plants in the Miyawaki forest in three sites had a higher growth rate. It is observed that the growth of the plants was faster than the usual field plant. When grown under optimum conditions, plant species from fertile, productive habitats have higher relative growth rates than species from less favorable environments (Lambers, 2008). Plant morphology determines the access to soil resources, a feature crucial for early growth in annual species (Alvarez-Flores *et al.*, 2014). Mixed plantations with many species planted densely lead to good growth because of the differences in crown spaces (Miyawaki, 1988). The forest improvement could effectively improve the soil quality, and the effect of the Miyawaki method is the best afforestation method compared to traditional afforestation methods (Guo, 2018). The Miyawaki technique in around 550 areas in Japan and Malaysia, Southeast Asia, Brazil, Chile, and a few zones of China was fruitful, permitting brisk natural rebuilding efforts of firmly corrupted zones (Goveanthan *et al.*, 2019).

PLATE 1: STUDY SITES



A: Kanakakunnu, **B:** Govt. High School, Chala **C:** Dept. of Botany, Kariavattom

Table 1. Categorization of plants based on their habit and growth pattern in site 1

Site 1: KANAKAKUNNU					
SI No.	Botanical Name	Common Name	Advantage (Medicinal/Ti mber/Fruit/Fl ower)	Type (Evergreen/d eciduous/per ennial)	Layer (Trees/canop y SubTrees/ Shrubs Herbs)
1	<i>Ceiba pentandra</i> (L.) Gaertn.	White cotton Trees Panji maram	Fruit	Deciduous	Trees
2	<i>Holoptelea integrifolia</i>	Indian elm, Aval	Medicinal	Deciduous	Trees
3	<i>Cassia fistula</i> L.	Indian Laburnum, Kanikkonna	Flower	Deciduous	Trees
4	<i>Thespesia populnea</i> (L.) Soland. ex Correa	Indian tulip Trees Poovarasu,	Medicinal	Evergreen	Sub Trees
5	<i>Azadirachta indica</i> A. Juss.	Neem, Ariyaveppu	Medicinal	Evergreen	Trees
6	<i>Artocarpus hirsutus</i> Lam.	Wild jack, Ayani	Fruit	Evergreen	Canopy
7	<i>Averrhoa bilimbi</i> L.	Bilimbi, Irumban puli	Fruit	Evergreen	Sub Trees
8	<i>Ficus carica</i> L.	Common fig	Medicinal	Deciduous	Trees
9	<i>Vitex negundo</i> L.	Common Chaste Trees, Karinochi	Medicinal	Deciduous	Shrubs/sub Trees
10	<i>Terminalia catappa</i> L.	Indian almond Trees, Badam	Fruit	Deciduous	Trees
11	<i>Tectona grandis</i> L.f.	Teak, Thekku	Timber	Deciduous	Trees
12	<i>Spathodea campanulata</i> Beauv.	African tulip tee, Phaauntanmaram	Timber	Deciduous	Trees
13	<i>Lennea coromandelica</i>	Indian ash Trees	Timber	Deciduous	Trees
14	<i>Albizzia lebbeck</i>	Vaga	Timber	Deciduous	Trees
15	<i>Theobroma cacao</i>	Cocoa Trees, Kokkoo	Fruit	Evergreen	Trees
16	<i>Morus alba</i> L.	Mulberry	Fruit	Deciduous	Trees
17	<i>Spondias pinnata</i> (L. f.) Kurz	Hog plum, Ambazham	Fruit	Deciduous	Trees

18	<i>Xanthophyllum arnottianum</i>	Madukka	Timber	Evergreen	Trees
19	<i>Carica papaya</i> L.	Papaya	Fruit	Evergreen	Sub Trees
20	<i>Annona reticulata</i> L.	Bullock's heart, Custard apple	Fruit	Evergreen	Trees
21	<i>Ixora</i> sp.	Chethi	Medicinal	Evergreen	Shrubs
22	<i>Clerodendrum inerme</i>	Wild jasmine	Medicinal	Evergreen	Shrubs
23	<i>Melaleuca</i> sp.		Medicinal	Evergreen	Shrubs
24	<i>Bauhinia acuminata</i> L.	Dwarf white Bauhinia, Mandaaram	Medicinal	Evergreen	Shrubs
25	<i>Ochlandra travancorica</i> (Bedd.) Gamble	Eera	Timber	Evergreen	Grass
26	<i>Plumbago zeylanica</i> L.	White plumbago Vellakoduveli	Medicinal	Perennial	Shrubs
27	<i>Bambusa bambos</i> (L.)Voss	Bamboo, Mula	Timber	Deciduous	Grass
28	<i>Ochlandra</i> sp.	Eera	Timber	Evergreen	Grass
29	<i>Adhatoda vasica</i> Nees	Adhatoda, Aadalodakam	Medicinal	Evergreen	Shrubs
30	<i>Canthium coromandelicum</i> (Burm.f.) Alston	Carray cheddie, Wild jessamine	Medicinal	Evergreen	Shrubs
31	<i>Asparagus racemosus</i> Willd.	Indian asparagus, Sathavari	Medicinal	Perennial	Herbs
32	<i>Commelina benghalensis</i> L.	Dayflower, Adukkavettilla	Medicinal	Perennial	Herbs
33	<i>Ludwigia</i> sp.	Swamp primrose	medicinal	Evergreen/pe rennial	Herbs
34	<i>Mirabilis Jalapa</i> L.	4 O'clock plant, Naalumani poovu	Medicinal	Perennial	Herbs
35	<i>Clitoria ternatea</i> L.	Butterfly bean, Sangu pushpam	Medicinal	Perennial	Herbs
36	<i>Muehlenbeckia platyclada</i>	Ribbon plant	Medicinal	Evergreen	Shrubs
37	<i>Rivina humilis</i> L.	Blood berry, Baby pepper,	Fruit	Perennial	Herbs

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Table 2. Categorization of plants based on their habit and growth pattern in site 2

Site.2: Government High School, Chala					
Sl No.	Botanical Name	Common Name	Advantage Medicinal/T imber/Fruit/ Flower	Type Evergreen /deciduous /perennial	Layer Trees/canopy/ Sub Trees/Shrubs Herbs
1	<i>Ficus carica</i> L.	Common fig	Medicinal	Deciduous	Trees
2	<i>Pithecellobium</i> sp.		Fruit	Deciduous	Trees
3	<i>Saraca asoca</i> (Roxb.)Willd.	Ashokam	Medicinal	Evergreen	Trees
4	<i>Hopea parviflora</i>	Thampakam	Timber	Evergreen	Trees
5	<i>Pongamia pinnata</i> (L.)Pierre	Indian beech, Pongam	Medicinal	Evergreen	Canopy
6	<i>Tamarindus indica</i> L.	Indian date, Tamarind, Puli	Fruit	Evergreen	Trees/Canopy
7	<i>Artocarpus heterophyllus</i> Lam.	Jack fruit Trees, plavu	Fruit	Evergreen	Trees
8	<i>Averrhoa bilimbi</i> L	Bilimbi, Iumban puli	Fruit	Evergreen	Sub Trees
9	<i>Terminalia cuneata</i>	White murdah	Fruit	Evergreen	Trees
10	<i>Pometia</i> sp	Fijian longan	Fruit	Evergreen	Trees
11	<i>Azadirachta indica</i> A. Juss.	Neem, Ariyaveppu	Medicinal	Evergreen	Trees
12	<i>Aegle marmelos</i> (L.) Correa	Wood apple, Koovalam	Medicinal	Deciduous	Trees
13	<i>Dalbergia latifolia</i> Roxb.	Indian rose wood	Timber	Deciduous	Trees
14	<i>Thespesia populnea</i> (L.) Soland. ex Correa	Indian tulip Trees, Poovarasu	Medicinal	Evergreen	Sub Trees
15	<i>Sesbania grandiflora</i> (L.) Pers.	West Indian pea, Agasthi	Medicinal	Deciduous	Trees
16	<i>Mangifera indica</i> L.	Mango Trees, Mavu	Fruit	Evergreen	Trees

17	<i>Phyllanthus acidus</i> (L.) Skeels	Star Gooseberry	Fruit	Deciduous	Sub Trees
18	<i>Senna tora</i> (L.) Roxb.	Foetid cassia, Thakara	Medicinal	Perennial	Shrubs
19	<i>Butea monosperma</i> (Lam.) Taub.	Flame of Forest, Chamatha	Medicinal	Deciduous	Trees
20	<i>Caesalpinia sappan</i> L.	Chappannam	Medicinal	Evergreen	Trees
21	<i>Tectona grandis</i> L.f.	Teak, Thekku	Timber	Deciduous	Trees
22	<i>Cinnamomum malabattrum</i> (Burm. f.) Presl	Vayana	Medicinal	Evergreen	Trees
23	<i>Nerium oleander</i> L.	Oleander, Arali	Medicinal	Evergreen	Shrubs
24	<i>Calophyllum inophyllum</i> L.	Oil nut Trees, Punna	Fruit	Evergreen	Trees
25	<i>Bauhinia racemosa</i> Lam.	Burmese silk orchid, Mandaram	Flower	Deciduous	Trees
26	<i>Garcinia</i> sp	Malabar tamarind, Kudmapuli	Fruit	Evergreen	Trees
27	<i>Terminalia catappa</i> L.	Indian almond Trees, Badam	Fruit	Deciduous	Trees
28	<i>Syzygium cumini</i> L.	Njaval	Fruit	Evergreen	Trees
29	<i>Elaeocarpus serratus</i>	Karakka	Fruit	Evergreen	Trees
30	<i>Morinda citrifolia</i> L., nom. cons.	Indian mulberry, Manjanathi	Medicinal	Evergreen	Trees
31	<i>Millingtonia hortensis</i>	Indian cork Trees	Medicinal	Deciduous	Trees
32	<i>Artocarpus hirsutus</i> Lam.	Wild jack, Ayani	Fruit	Evergreen	Canopy
33	<i>Dillenia indica</i> L.	Elephant apple	Fruit	Evergreen	Trees
34	<i>Olea dioica</i> Roxb.	Rose sandal wood	Medicinal	Semi evergreen	Trees
35	<i>Vitex negundo</i> L.	Common Chaste Trees, Karinochi	Medicinal	Deciduous	Trees
36	<i>Clausena dentata</i>		Medicinal	Evergreen	Trees
37	<i>Persea macrantha</i> (Nees)	Kulamavu	Fruit/medicinal	Evergreen	Trees
38	<i>Strychnos nux-vomica</i>	Kaanjiram	Medicinal	Evergreen	Trees

39	<i>Justicia gendarussa</i> Burm. fil.	Water willow	Medicinal	Deciduous	Shrubs
40	<i>Calotropis gigantea</i> (L.) W. T. Aiton	Giant milk weed, Yerikku	Medicinal	Evergreen	Shrubs
41	<i>Eugenia uniflora</i> L.	Brazil cherry	Fruit	Evergreen	Shrubs
42	<i>Mussaenda</i> sp.			Evergreen	Shrubs
43	<i>Punica granatum</i> L.	Pomgranate, Mathalam	Fruit	Deciduous	Sub Trees
44	<i>Murraya paniculata</i> (L.) Jacq.	Orange Jessamine Maramulla	Medicinal	Evergreen	Shrubs
45	<i>Lawsonia inermis</i> L.	Henna, Milanji	Medicinal	Evergreen	Shrubs
46	<i>Morus alba</i> L.	Mulberry	Fruit	Deciduous	Trees
47	<i>Eleutheranthera ruderalis</i> (Sw.) Sch. -Bip	Pakaka	Medicinal	Perennial	Herbs
48	<i>Acalypha indica</i> L.	Indian acalypha, Kuppameni	Medicinal	Perennial	Herbs
49	<i>Cleome burmannii</i> Wight & Arn.	Kattukadugu	Medicinal	Perennial	Herbs
50	<i>Amaranthus</i> sp.		Medicinal	Perennial	Herbs
51	<i>Commelina benghalensis</i> L.	Dayflower, Adukkavettala	Medicinal	Perennial	Herbs
52	<i>Euphorbia hirta</i> L.	Snake head, Nilappala	Medicinal	Perennial	Herbs
53	<i>Synedrella nodiflora</i> (L.) Gaertn.	Cinderella Weed, Muriyan pacha	Medicinal	Perennial	Herbs
54	<i>Plumbago zeylanica</i> L.	White plumbago Vellakoduveli	Medicinal	Perennial	Shrubs
55	<i>Tridax procumbens</i> L.	Coat button, Odiyan cheera	Medicinal	Perennial	Herbs
56	<i>Boerhavia diffusa</i> L.	Common hog weed, Thazhuthama	Medicinal	Perennial	Herbs
57	<i>Ocimum tenuiflorum</i>	Thulasi	Medicinal	Perennial	Herbs
58	<i>Coccinia grandis</i>	Ivygourd	Fruit vegetable	Perennial	Herbs

Table 3. Categorization of plants based on their habit and growth pattern in site 3

Site 3: Department of Botany, Kariavattom					
Sl No.	Botanical Name	Common Name	Advantage Medicinal/T imber/Fruit/ Flower	Type Evergreen /deciduous /perennial	Layer Trees/canopy/ Sub Trees/Shrubs Herbs
1.	<i>Saraca asoca</i>	Ashokam	Medicinal	Evergreen	Trees
2.	<i>Alstonia</i> sp.	Devil Trees	Medicinal	Evergreen	Trees
3.	<i>Tamarindus indica</i>	Tamarind, puli	Fruit	Evergreen	Trees
4.	<i>Artocarpus heterophyllus</i> Lam.	Jack fruit Trees, plavu	Fruit	Evergreen	Trees
5.	<i>Dalbergia latifolia</i> Roxb.	Indian rose wood	Timber	Deciduous	Trees
6.	<i>Thespesia populnea</i> (L.) Soland. ex Correa	Indian tulip Trees Poovarasu,	Medicinal	Evergreen	Sub Trees
7.	<i>Cinnamomum malabattrum</i> (Burm. f.) Presl	Vayana	Medicinal	Evergreen	Trees
8.	<i>Psidium guajava</i>	Guava	Fruit	Evergreen	Trees
9.	<i>Pongamia pinnata</i> (L.) Pierre	Indian beech, Pongam	Medicinal	Evergreen	Canopy
10.	<i>Averrhoa bilimbi</i> L	Bilimbi, Irumban puli	Fruit	Evergreen	Sub Trees
11.	<i>Polyalthia longifolia</i>	Cemetery Trees	Timber	Evergreen	Trees
12.	<i>Syzygium cumini</i> L.	Njaval	Fruit	Evergreen	Trees
13.	<i>Azadirachta indica</i> A. Juss.	Neem, Ariyaveppu	Medicinal	Evergreen	Trees
14.	<i>Annona reticulata</i>	Soursop, mulluatha	Fruit	Evergreen	Trees
15.	<i>Calophyllum inophyllum</i> L.	Oil nut Trees, Punna	Fruit	Evergreen	Trees
16.	<i>Plumeria rubra</i>	Frangipani	Flower	Deciduous	Trees
17.	<i>Madhuca indica</i>		Medicinal	Evergreen	Trees
18.	<i>Syzygium cumini</i>	Njaval	Fruit	Evergreen	Trees
19.	<i>Butea monosperma</i> (Lam.) Taub.	Flame of Forest, Chamatha	Medicinal	Deciduous	Trees

20.	<i>Averrhoa carambola</i>	Star apple	Fruit	Evergreen	Sub Trees
21.	<i>Garcinia</i> sp.	Malabar tamarind, Kudampuli	Fruit	Evergreen	Trees
22.	<i>Aegle marmelos</i> (L.) Correa	Wood apple, Koovalam	Medicinal	Deciduous	Trees
23.	<i>Cassia fistula</i> L.	Indian Laburnum, Kanikkonna	Flower	Deciduous	Trees
24.	<i>Dillenia indica</i> L.	Elephant apple	Fruit	Evergreen	Trees
25.	<i>Phyllanthus emblica</i>	Indian gooseberry	Fruit	Deciduous	Trees
26.	<i>Limonia acidissima</i>	Wood apple	Fruit	Deciduous	Trees
27.	<i>Melia azedarach</i>	Pride of India	Medicinal	Evergreen	Sub Trees
28.	<i>Putranjiva roxburghii</i>	Lucky bean Trees	Medicinal	Evergreen	Trees
29.	<i>Annona muricata</i>	Prickly custard	Fruit	Evergreen	Trees
30.	<i>Morus alba</i> L.	Mulberry	Fruit	Deciduous	Trees
31.	<i>Eugenia uniflora</i> L.	Brazil cherry	Fruit	Evergreen	Shrubs
32.	<i>Justicia adhatoda</i>	Adalodakam	Medicinal	Evergreen	Shrubs
33.	<i>Hibiscus rosa-sinensis</i>	China rose	Flower	Evergreen	Shrubs
34.	<i>Nerium oleander</i> L.	Oleander, Arali	Medicinal	Evergreen	Shrubs
35.	<i>Lawsonia inermis</i> L.	Henna, Milanji	Medicinal	Evergreen	Shrubs
36.	<i>Ixora coccinea</i>	Chethi	Medicinal	Evergreen	Shrubs
37.	<i>Coleus amboinicus</i>	Panikkurkka	Medicinal	Perennial	Herbs
38.	<i>Pogostemon heyneanus</i>	Indian Patchouli	Medicinal	Evergreen	Shrubs

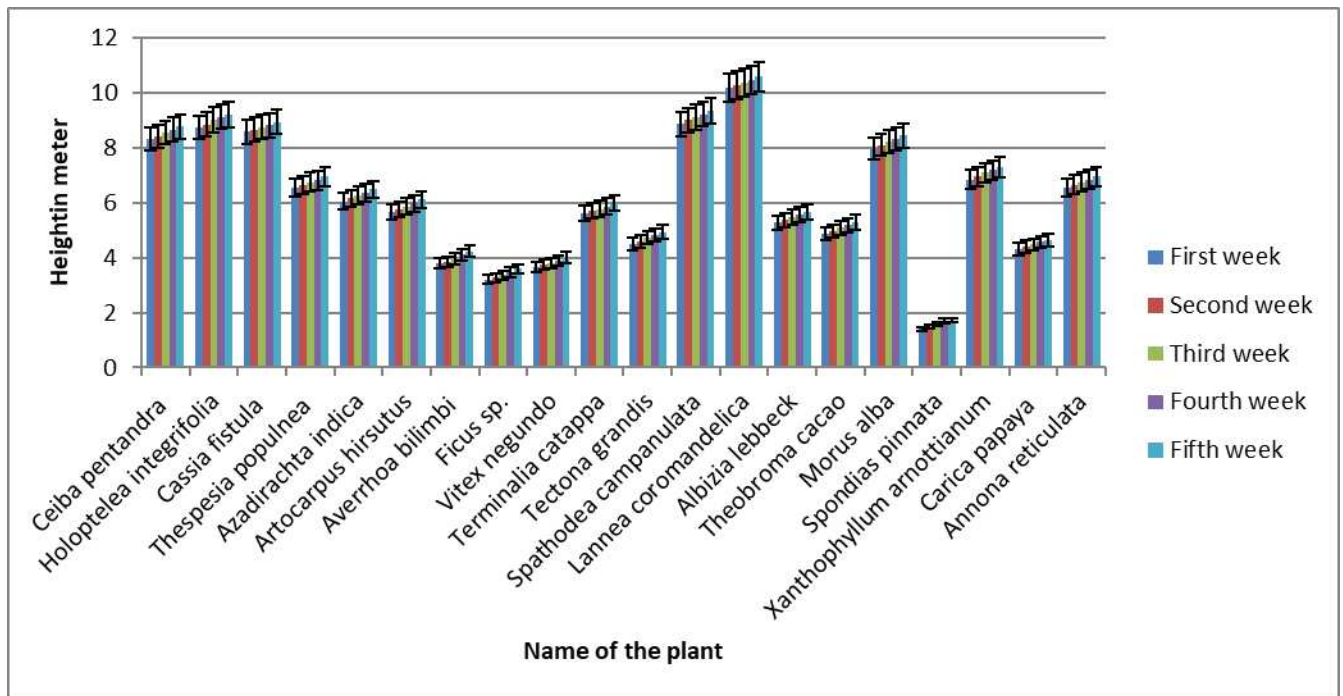


Figure 1. Height of the Trees in Site 1 (Kanakakunnu)

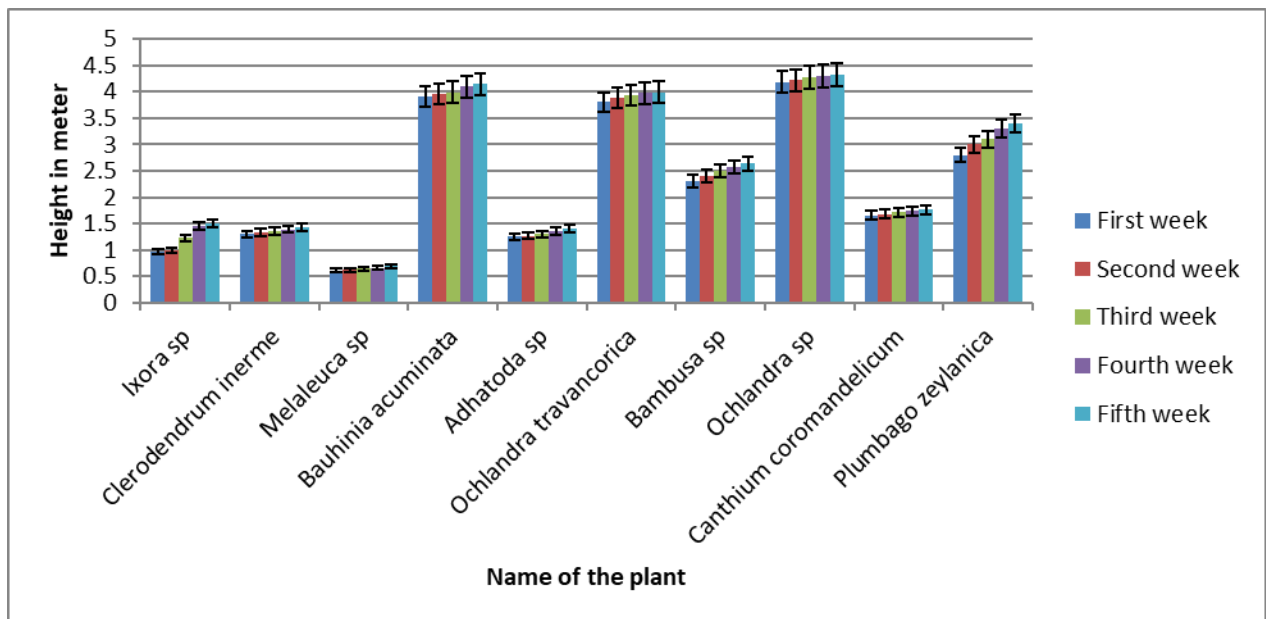


Figure 2. Height of the Shrubs in Site 1 (Kanakakunnu)

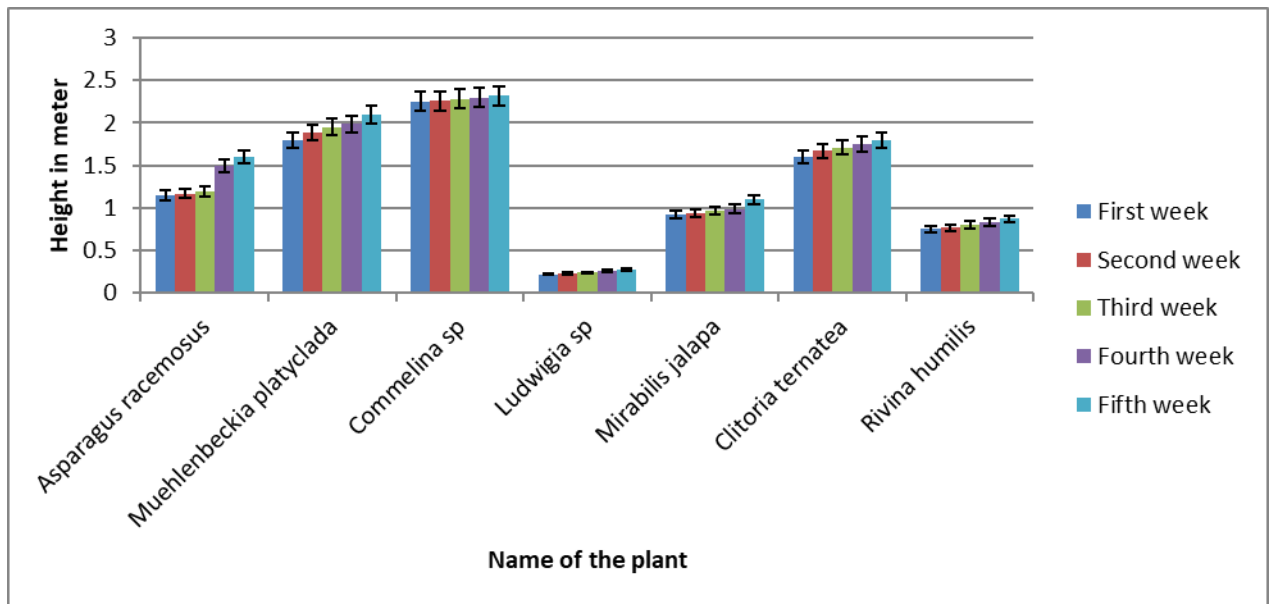


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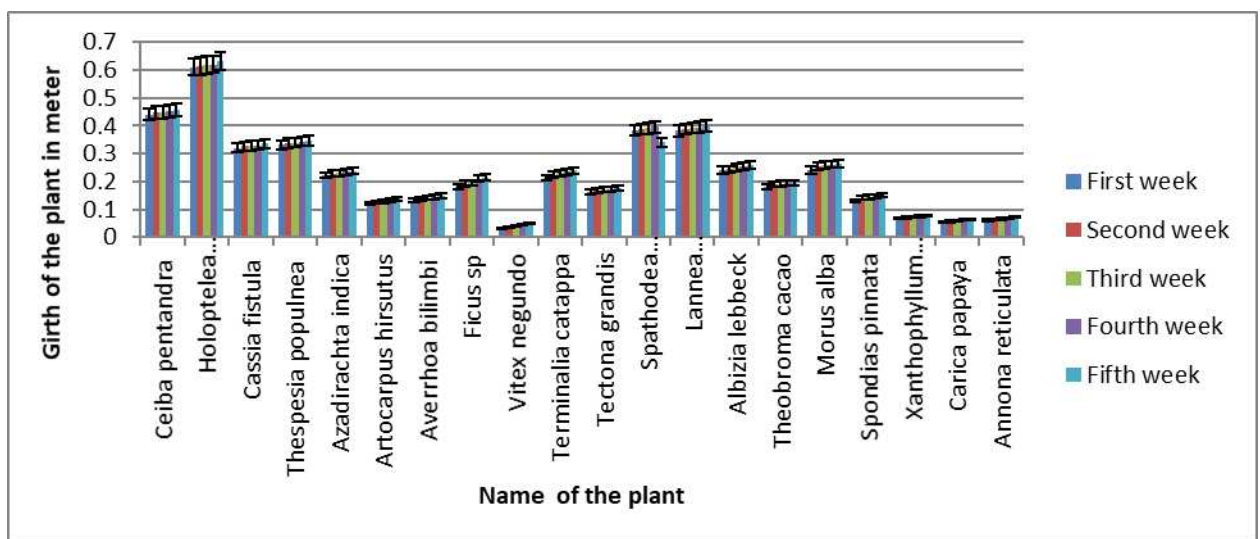


Figure 4. Girth of the Trees in Site 1 (Kanakakunnu)

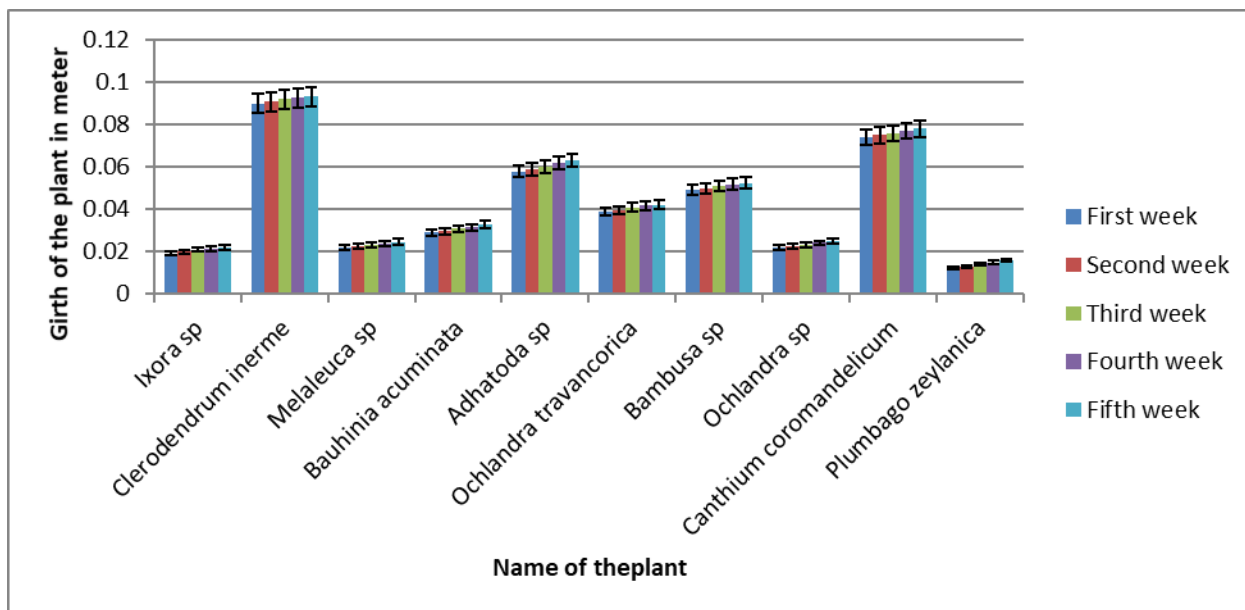


Figure 5. Girth of the Shrubs in Site 1 (Kanakakunnu)

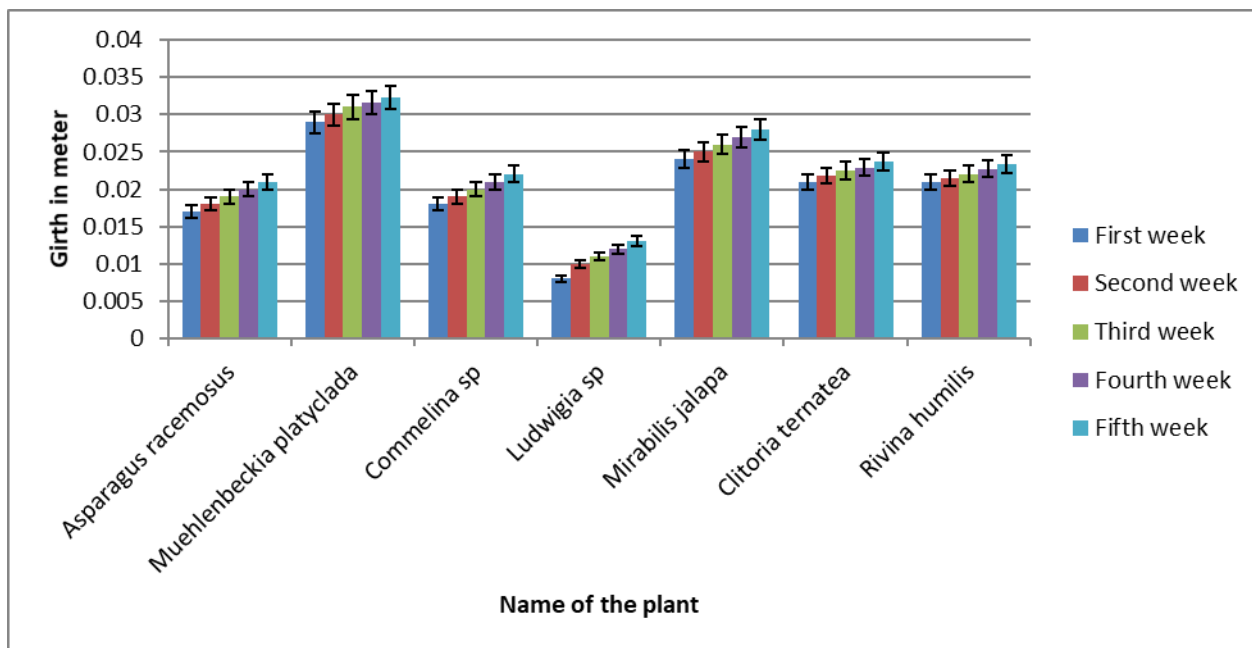


Figure 6. Girth of the Herbs in Site 1 (Kanakakunnu)

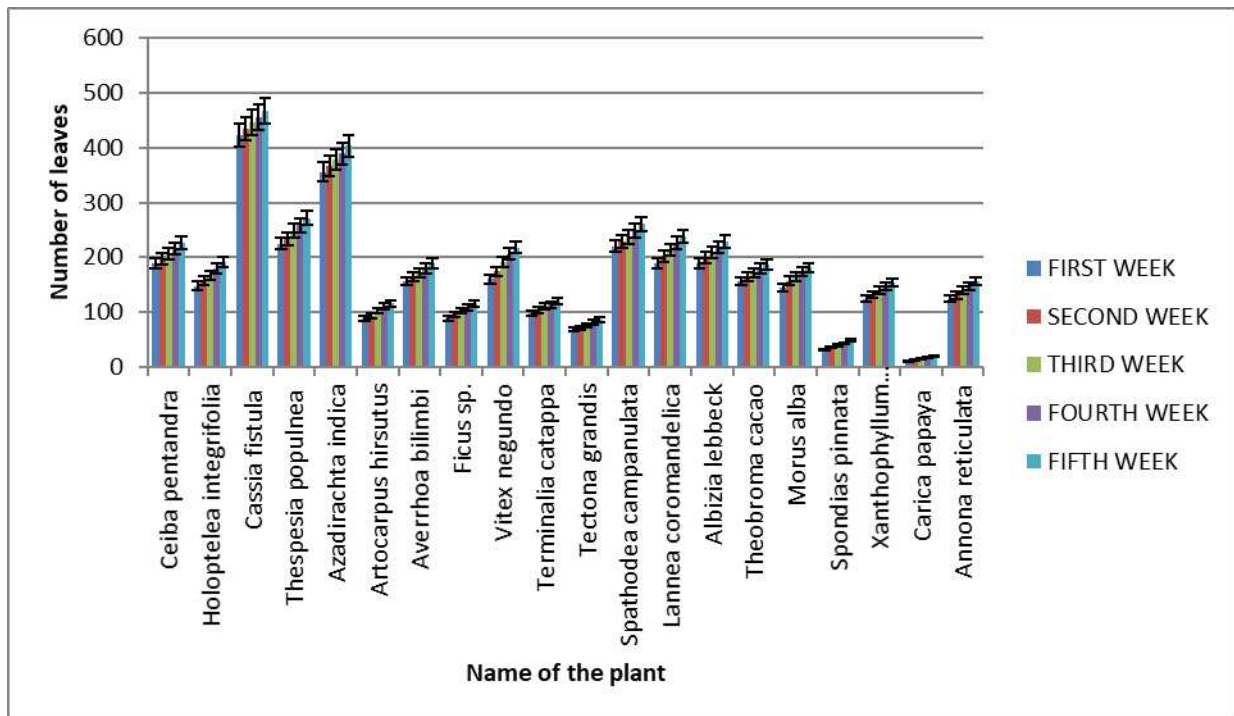


Figure 7. Number of leaves in Trees in Site 1 (Kanakakunnu)

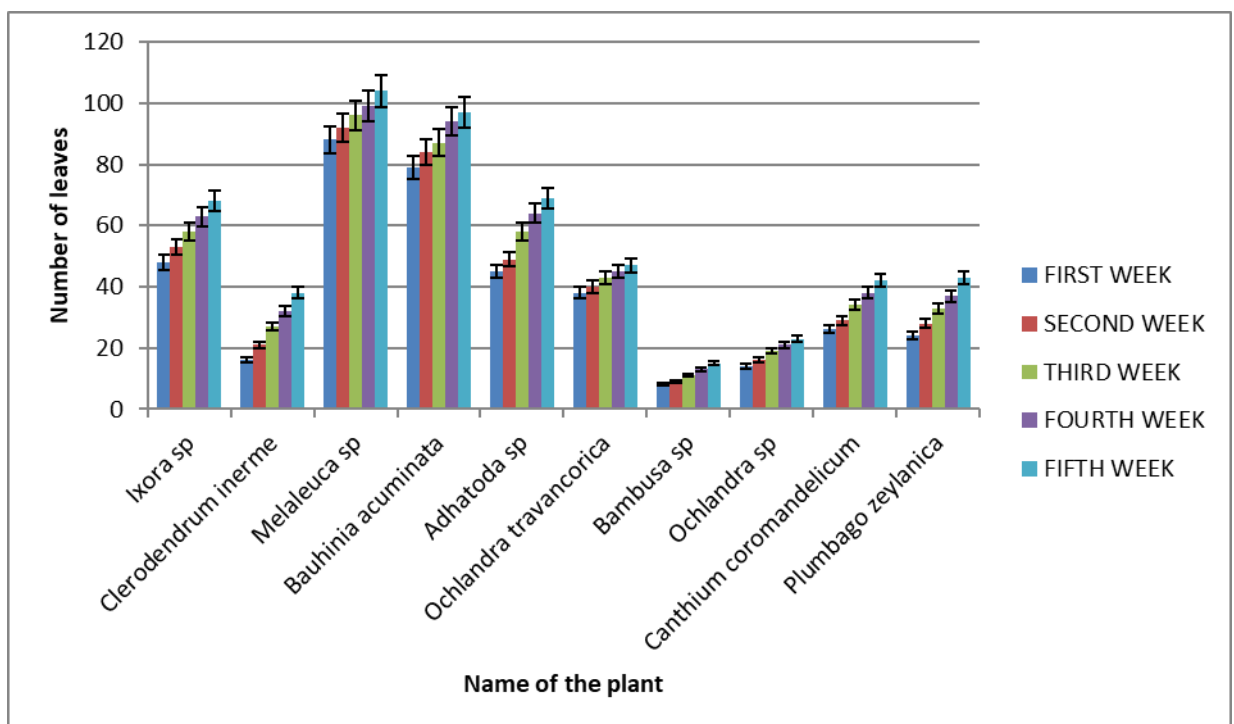


Figure 8. Number of leaves in Shrubs in Site 1 (Kanakakunnu)

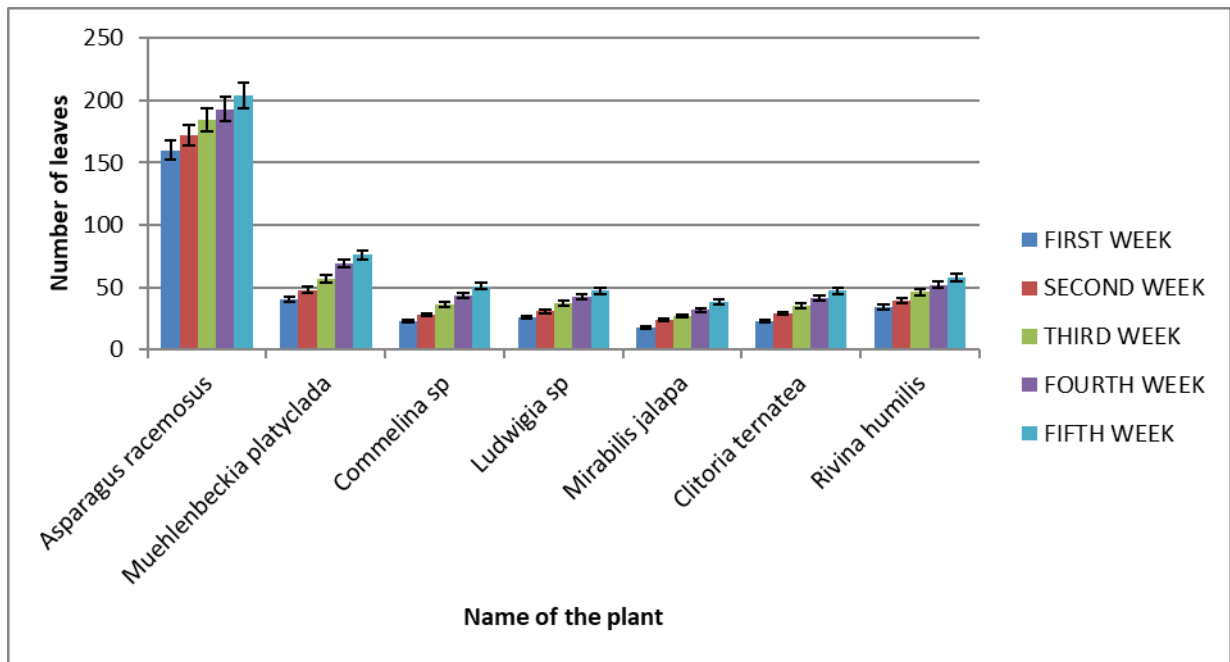


Figure 9. Number of leaves in Herbs in Site 1 (Kanakakunnu)

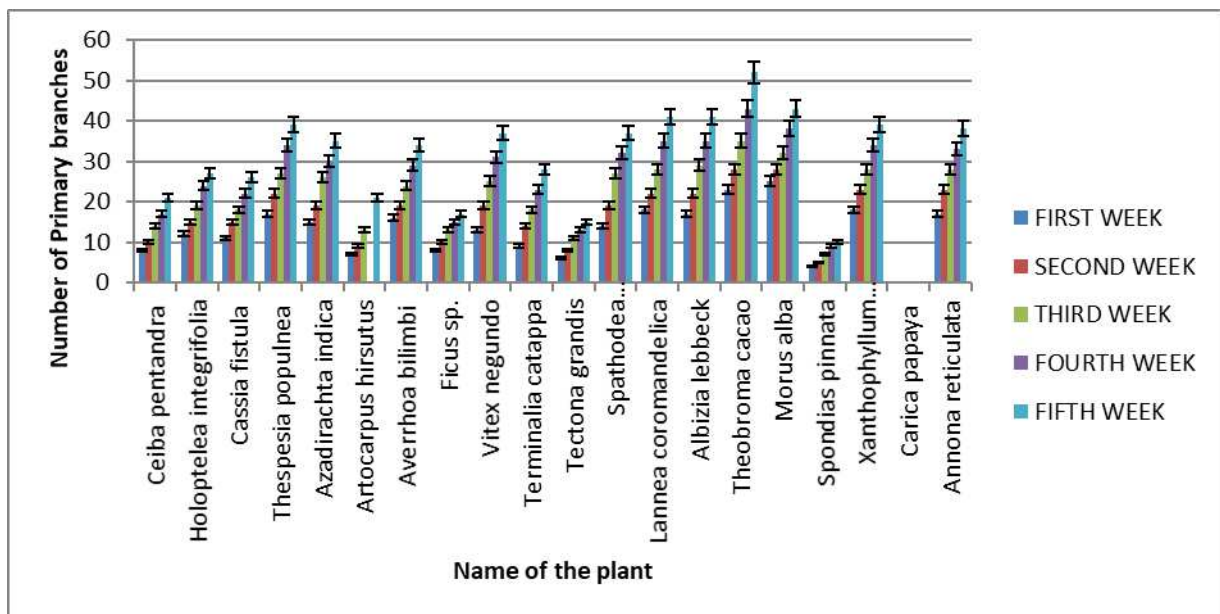


Figure 10. Number of Primary branches in Trees in Site 1 (Kanakakunnu)

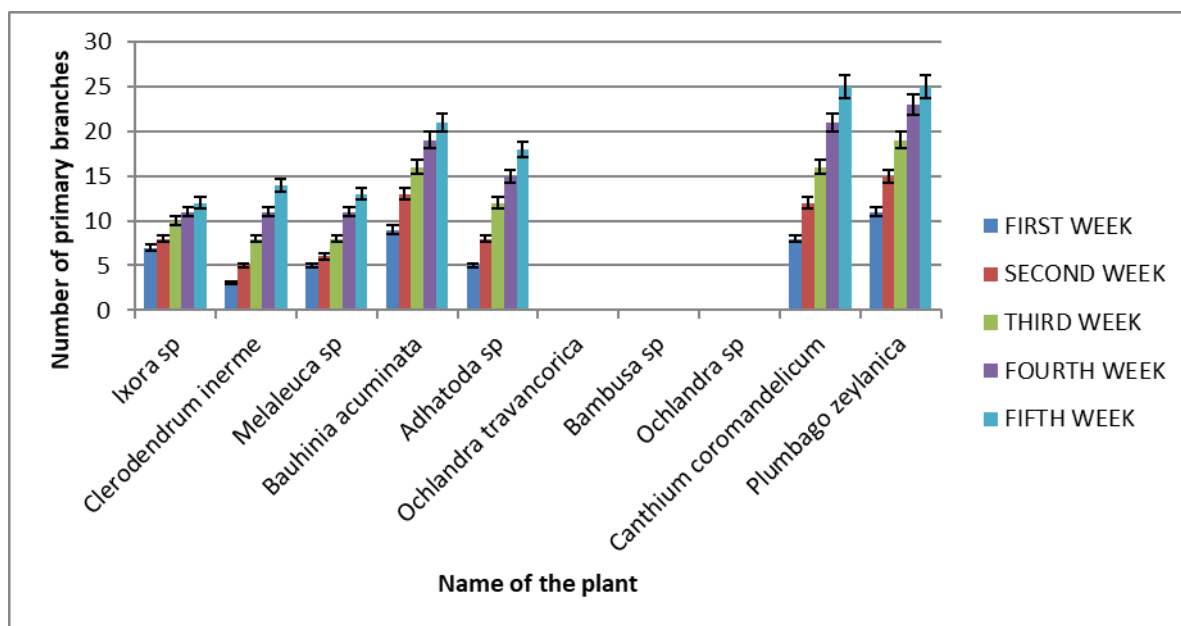


Figure 11. Number of Primary branches in Shrubs in Site 1 (Kanakakunnu)

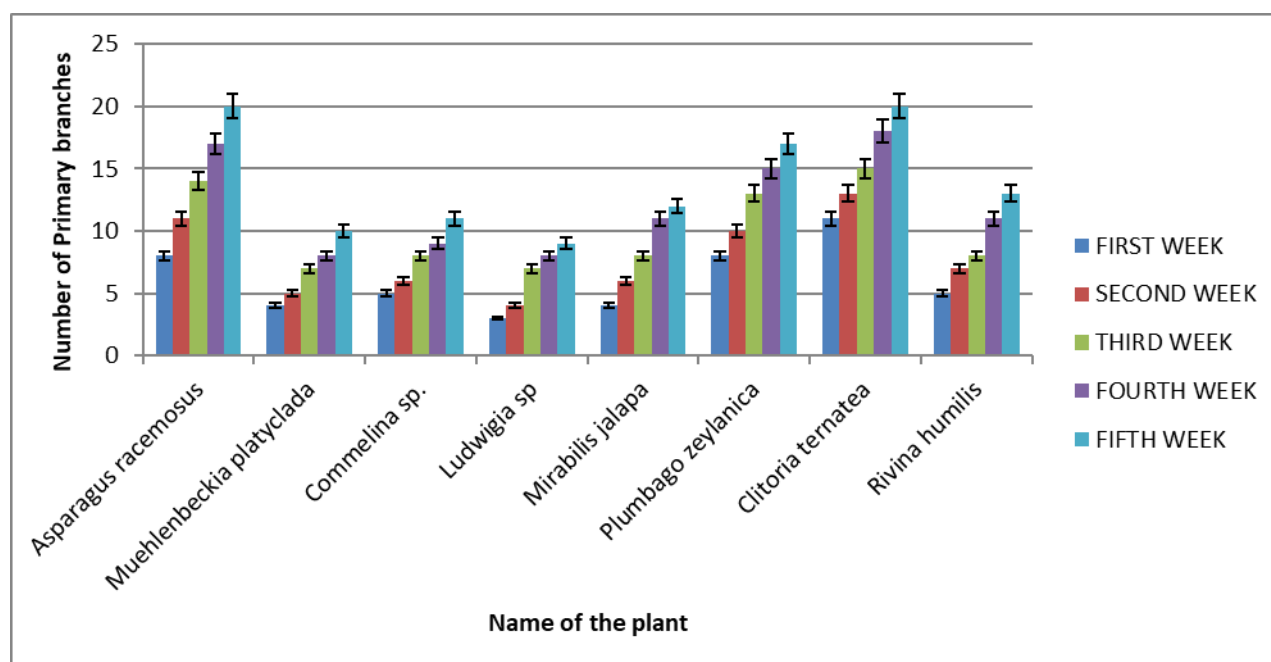


Figure 12. Number of Primary branches in Herbs in Site 1 (Kanakakunnu)

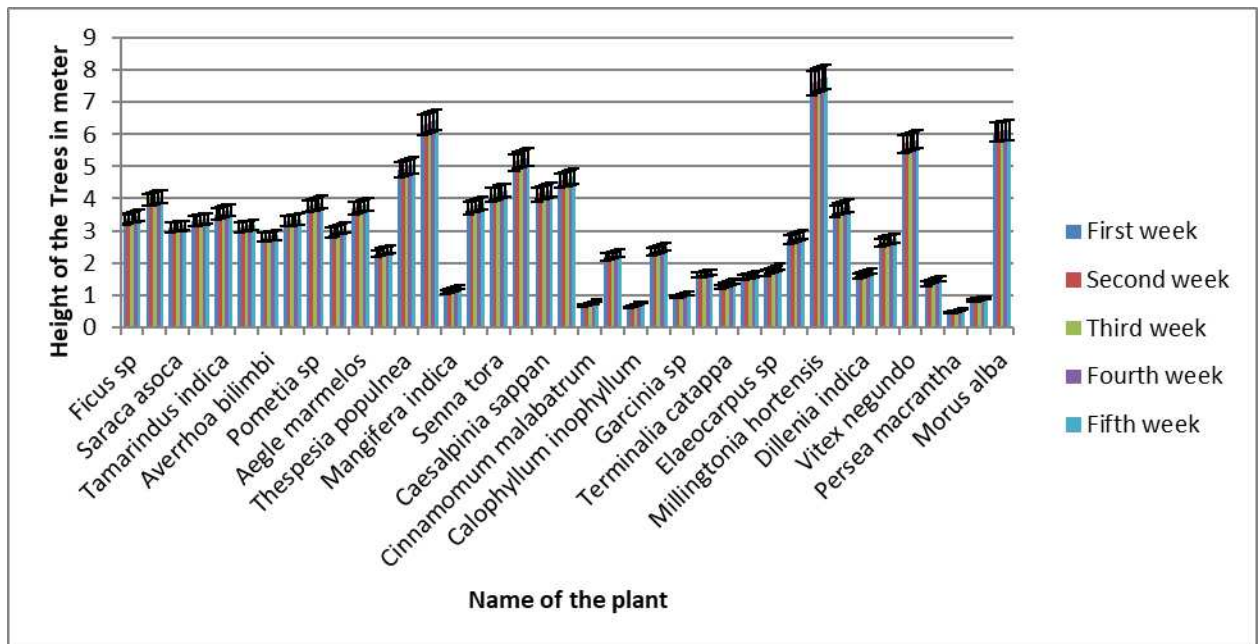


Figure 13. Height of the Trees in Site 2 (Govt. High School, Chala)

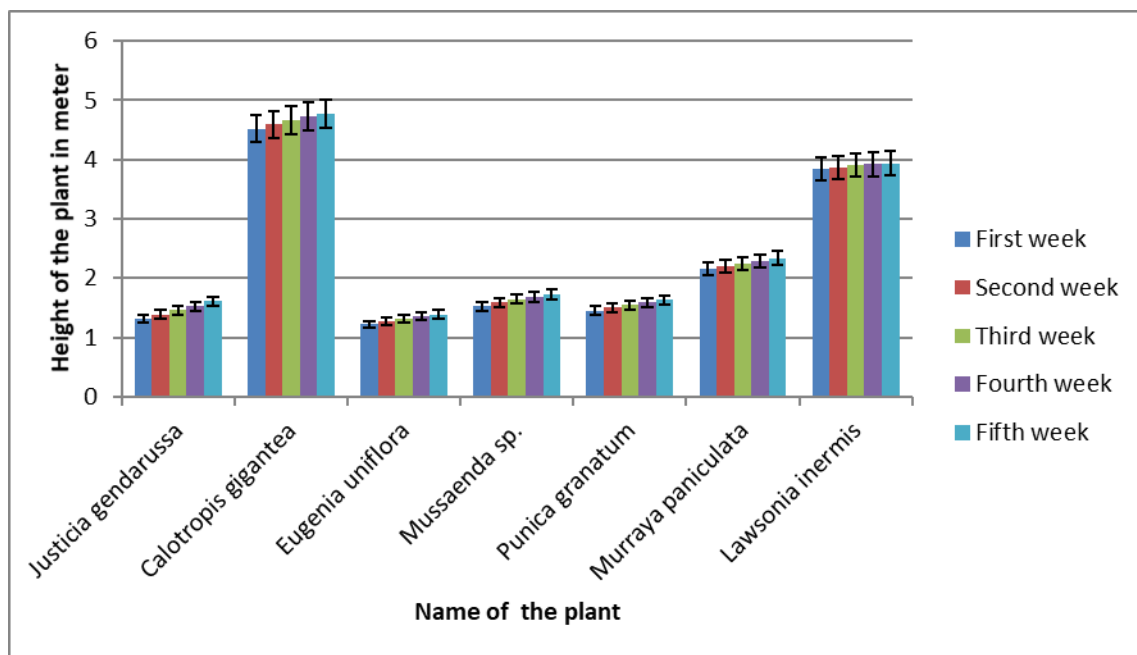


Figure 14. Height of the Shrubs in Site 2 (Govt. High School, Chala)

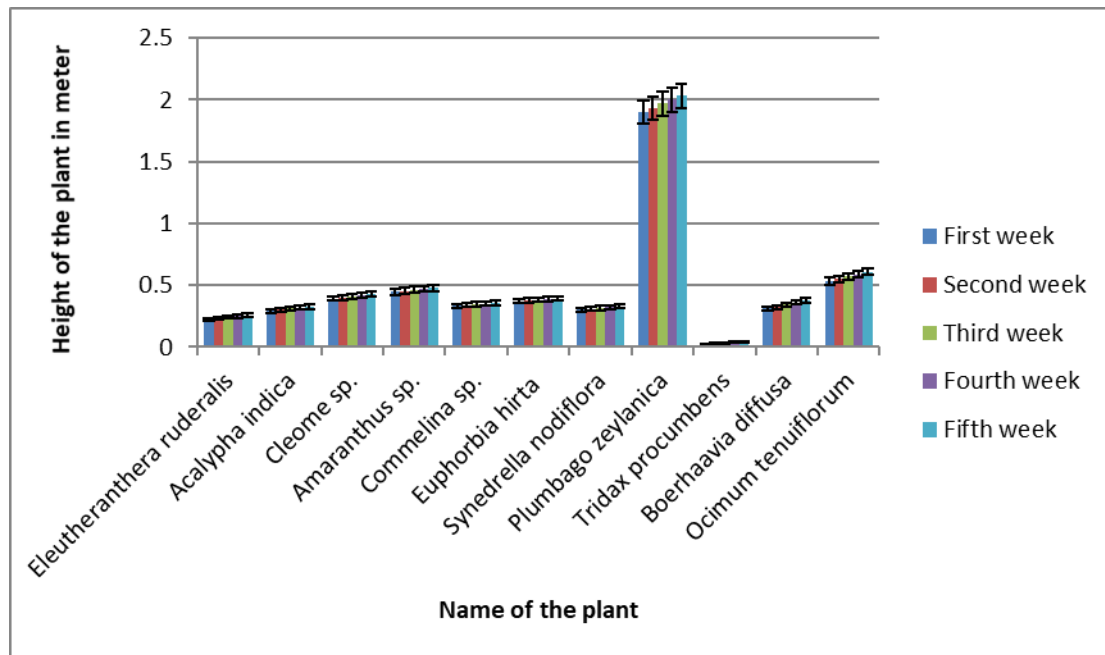


Figure 15. Height of the Herbs in Site 2 (Govt. High School, Chala)

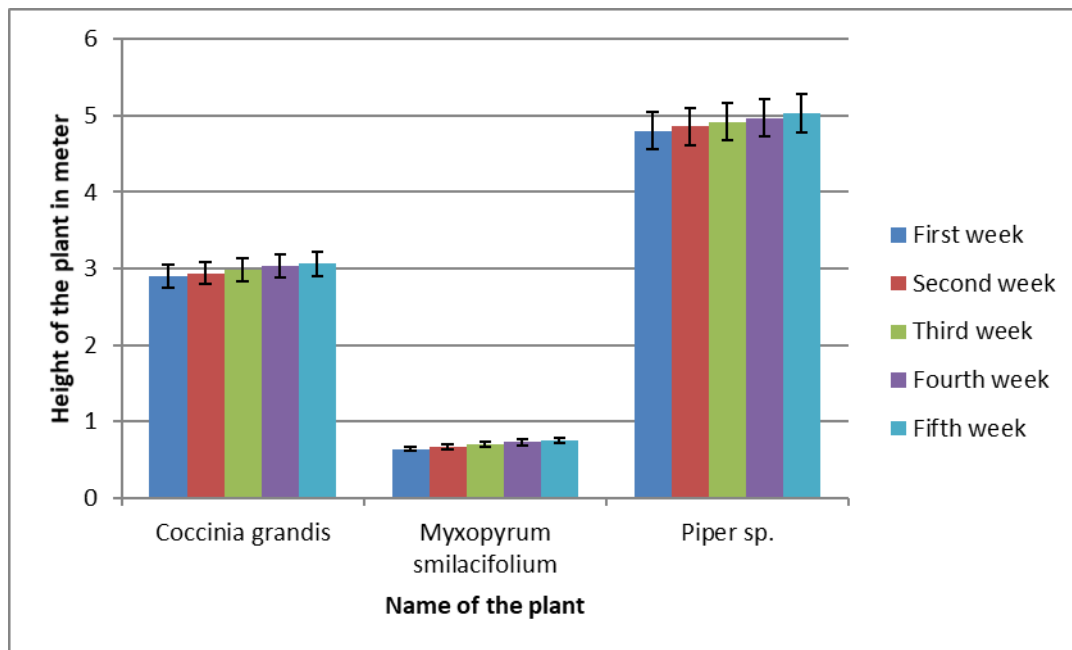


Figure 16. Height of the Climbers in Site 2 (Govt. High School, Chala)

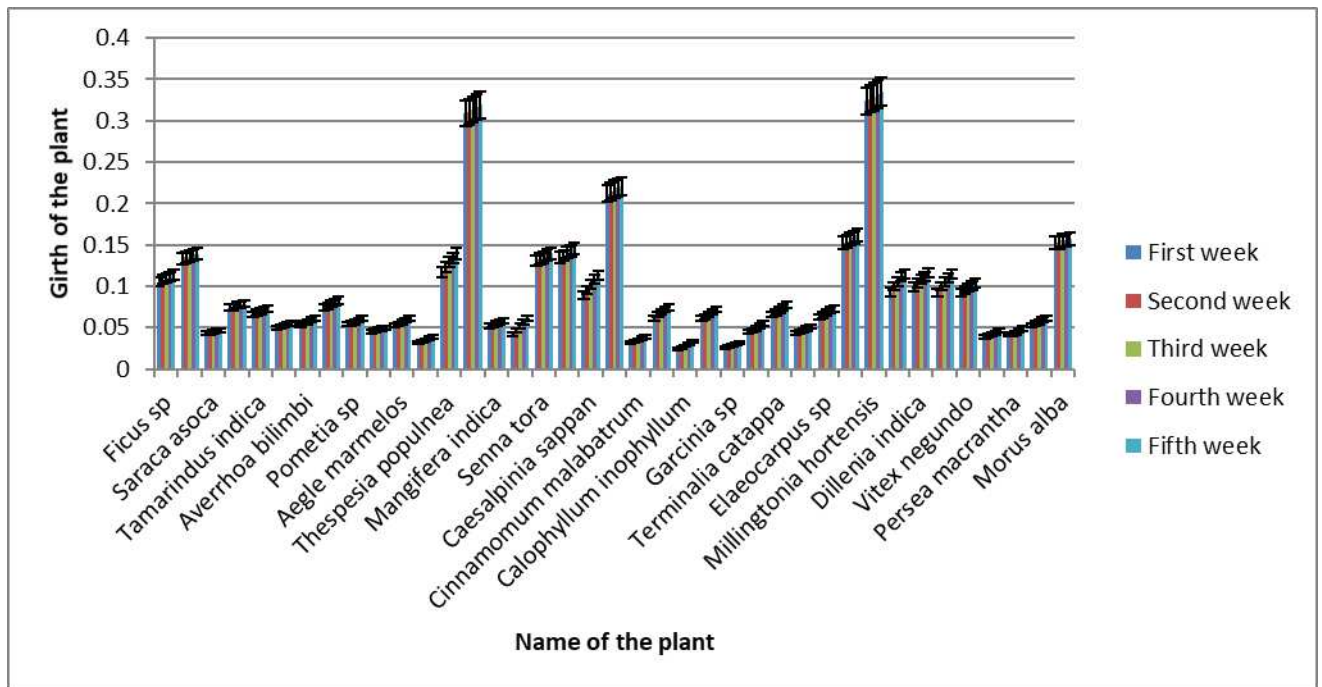


Figure 17. Girth of the Trees in Site 2 (Govt. High School, Chala)

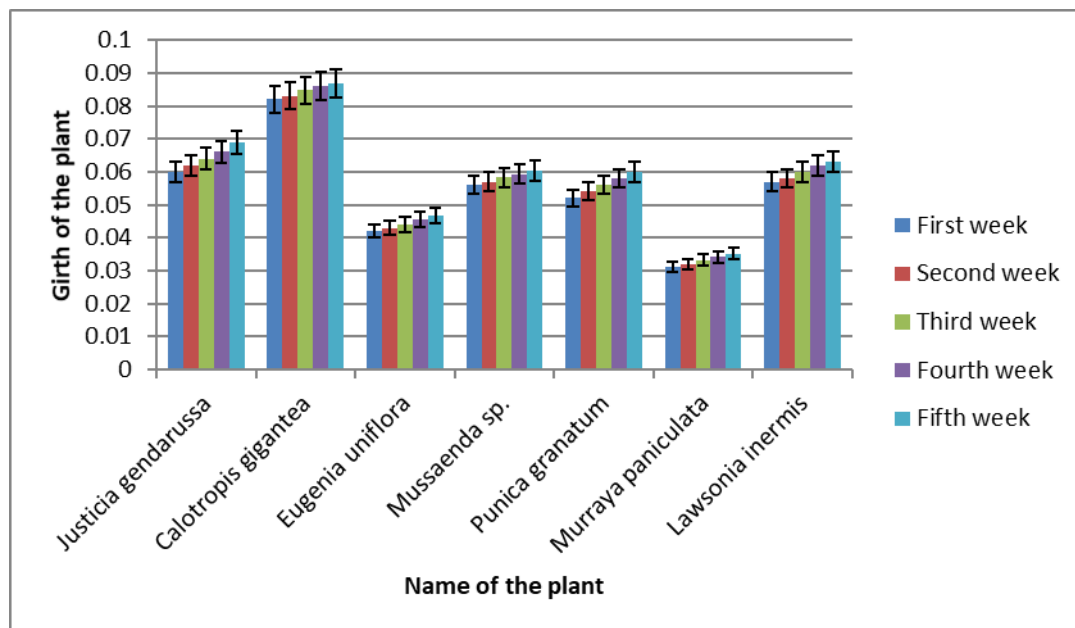


Figure 18. Girth of Shrubs in Site 2 (Govt. High School, Chala)

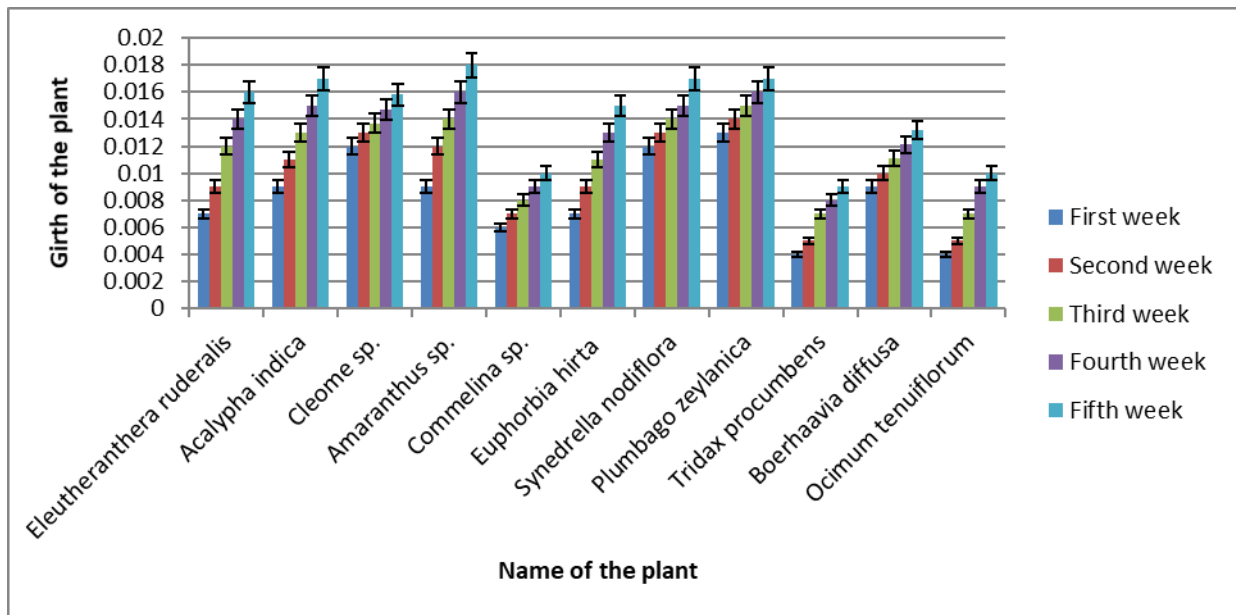


Figure 19. Girth of Herbs in Site 2 (Govt. High School, Chala)

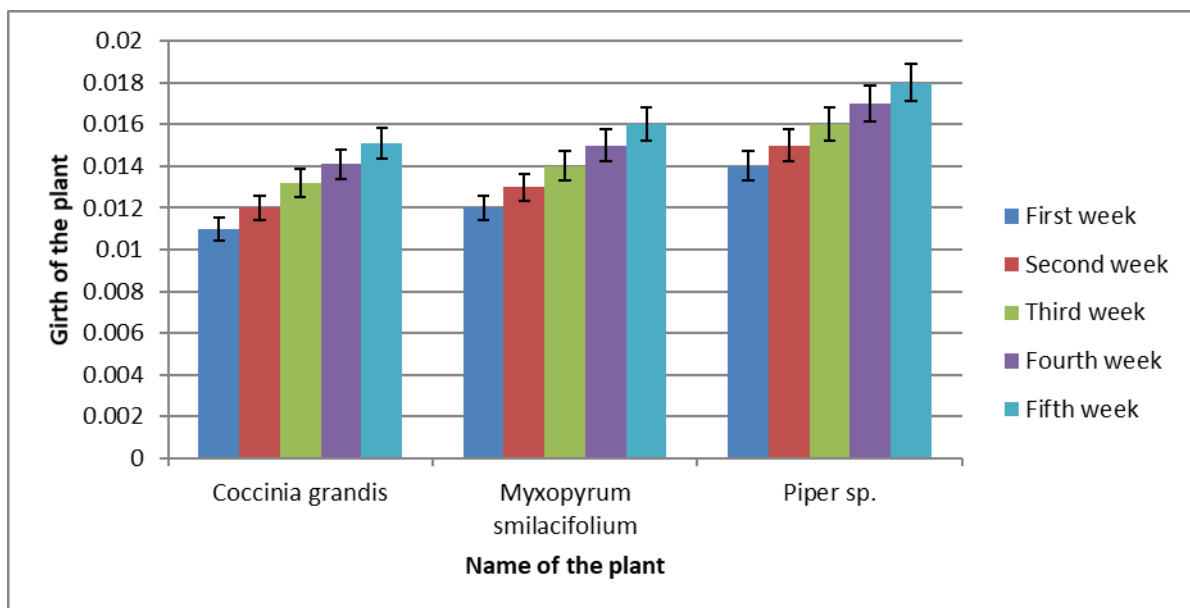


Figure 20. Girth of Climber in Site 2 (Govt. High School, Chala)

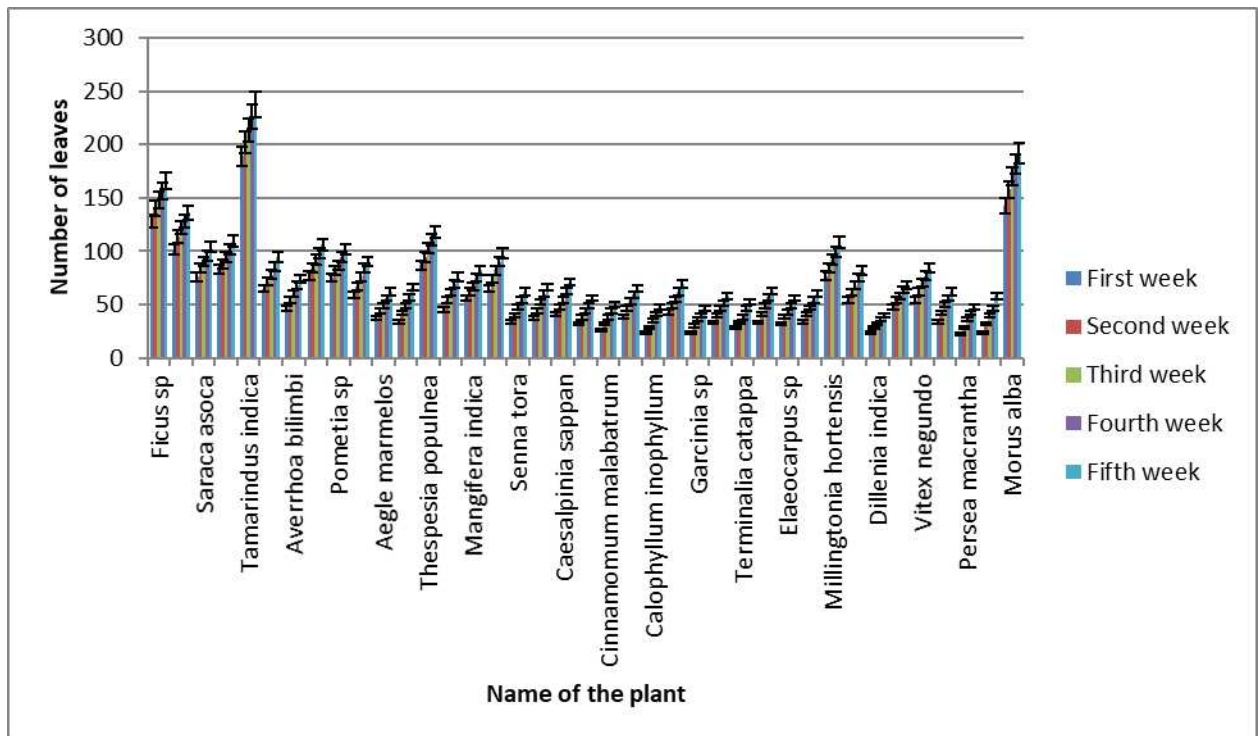


Figure 21. Number of leaves in Trees in Site 2 (Govt. High School, Chala)

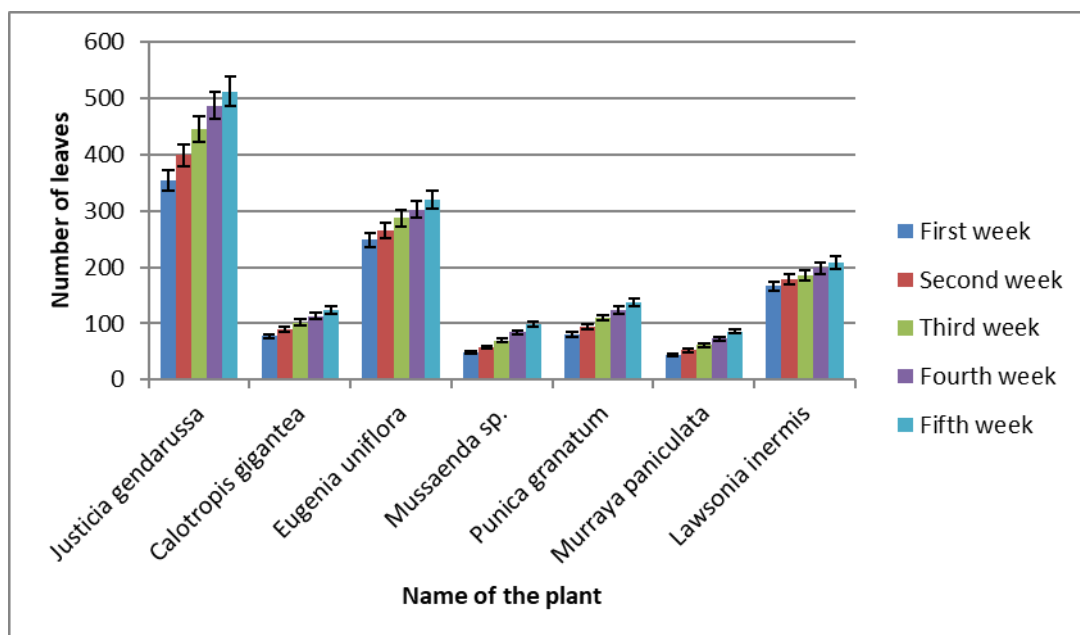


Figure 22. Number of leaves in Shrubs in Site 2 (Govt. High School, Chala)

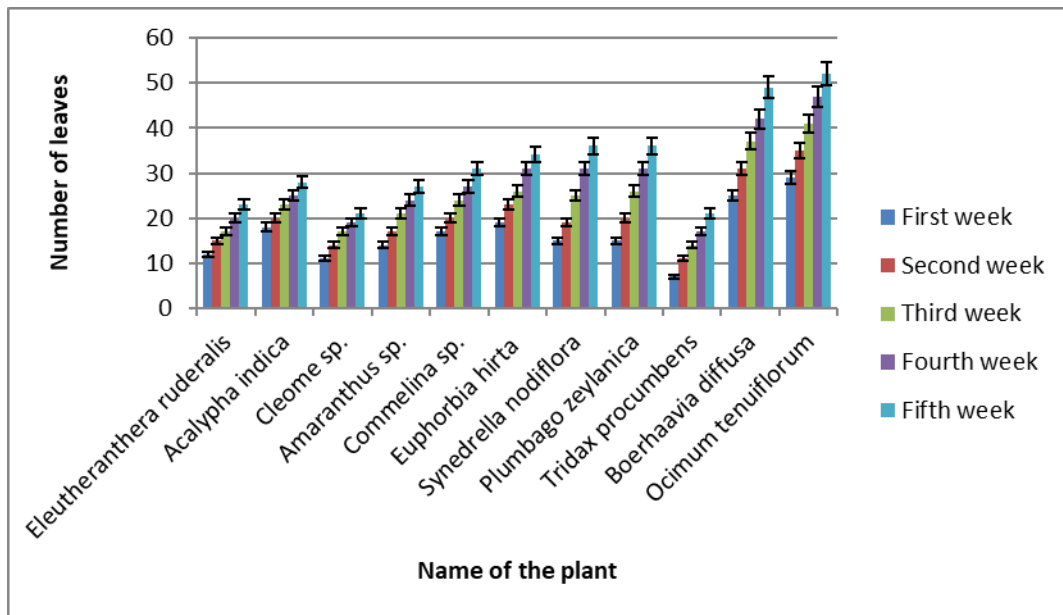


Figure 23. Number of leaves in Herbs in Site 2 (Govt. High School, Chala)

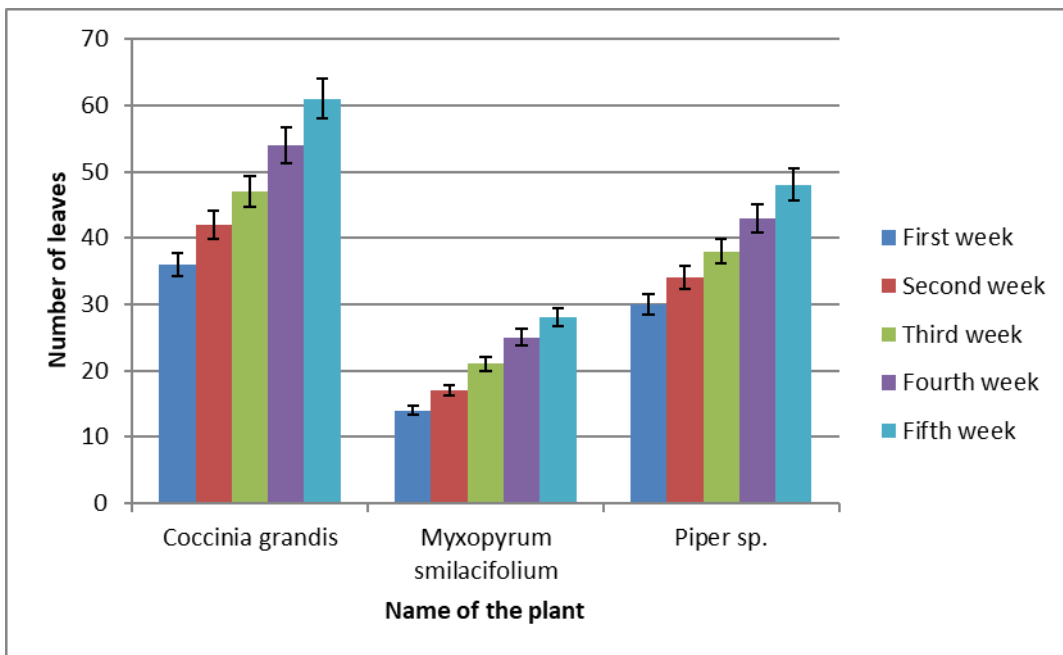


Figure 24. Number of leaves in Climber in Site 2 (Govt. High School, Chala)

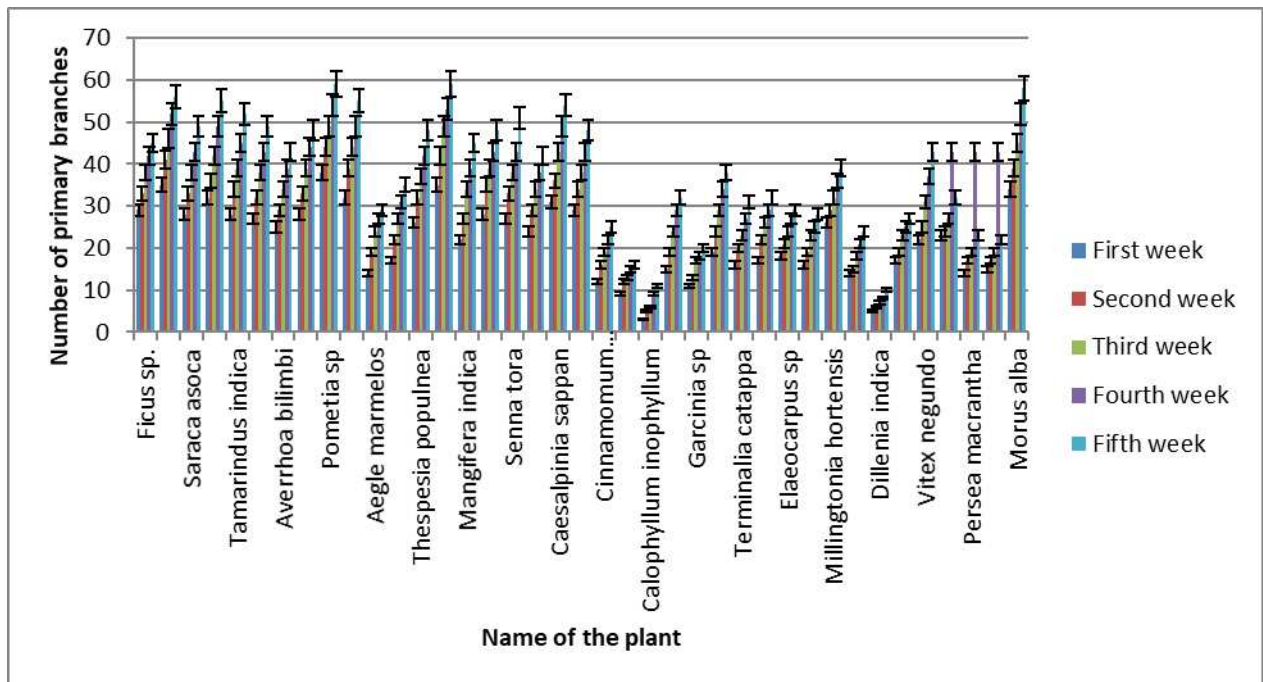


Figure 25. Number of Primary branches in Trees in Site 2 (Govt. High School, Chala)

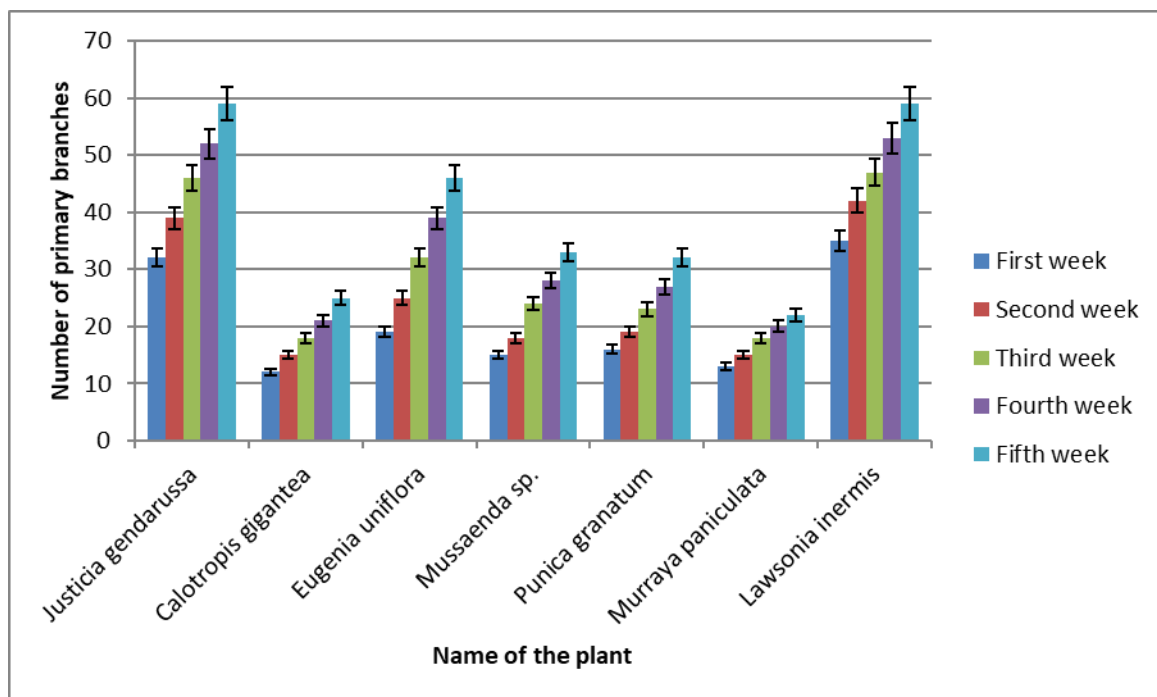


Figure 26. Number of Primary branches in Shrubs in Site 2 (Govt. High School, Chala)

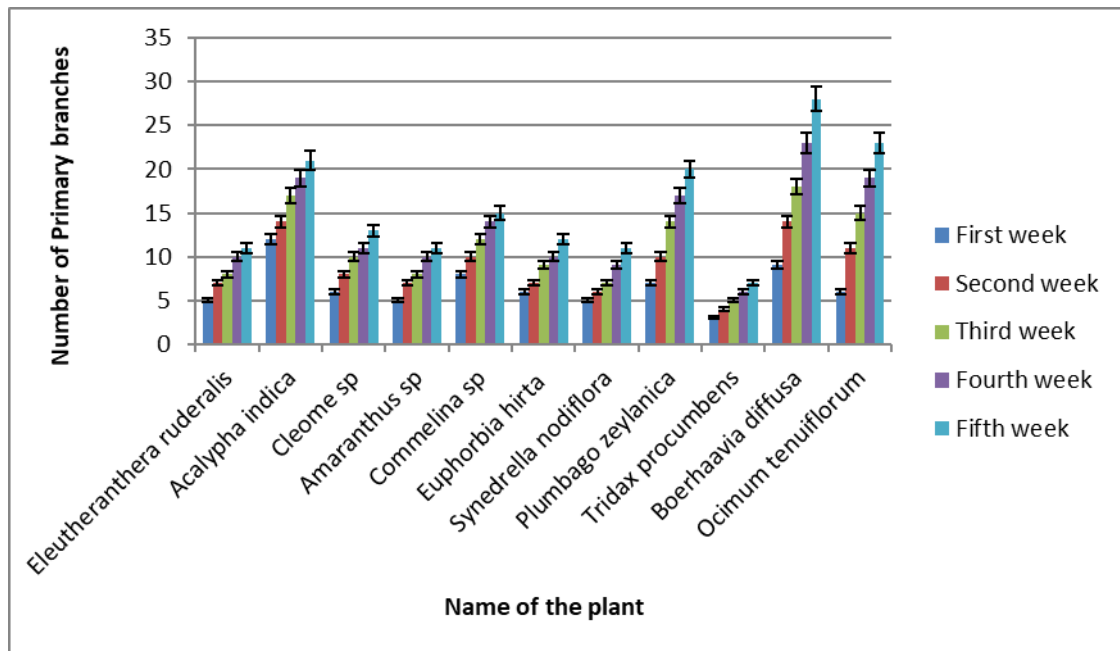


Figure 27. Number of Primary branches in Herbs in Site 2 (Govt. High School, Chala)

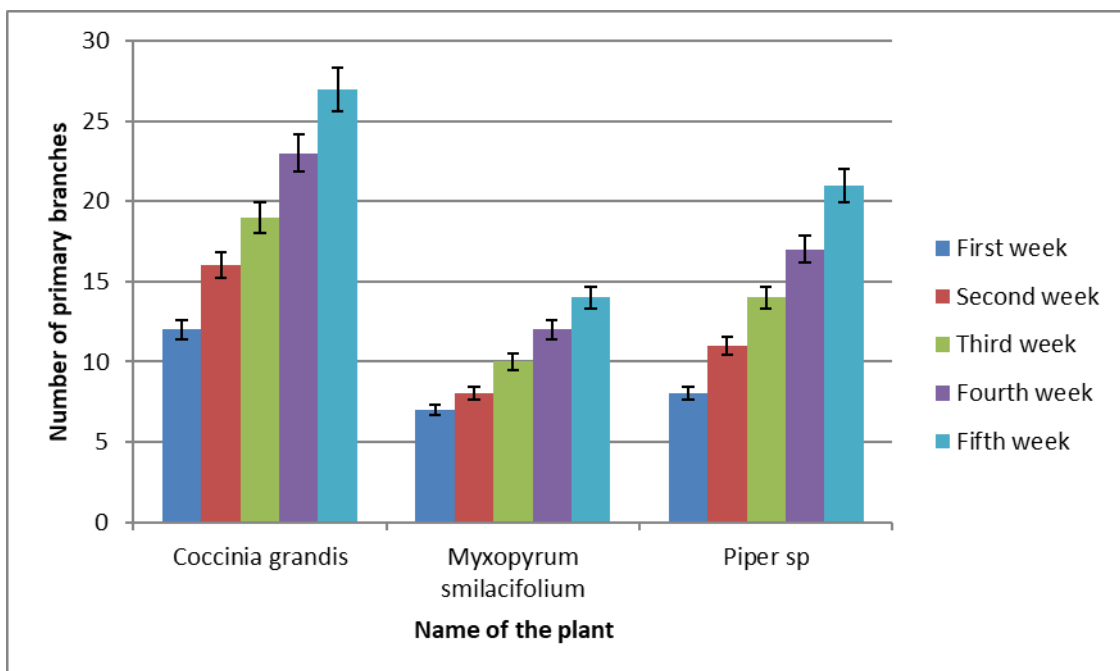


Figure 28. Number of Primary branches in Climbers in Site 2 (Govt. High School, Chala)

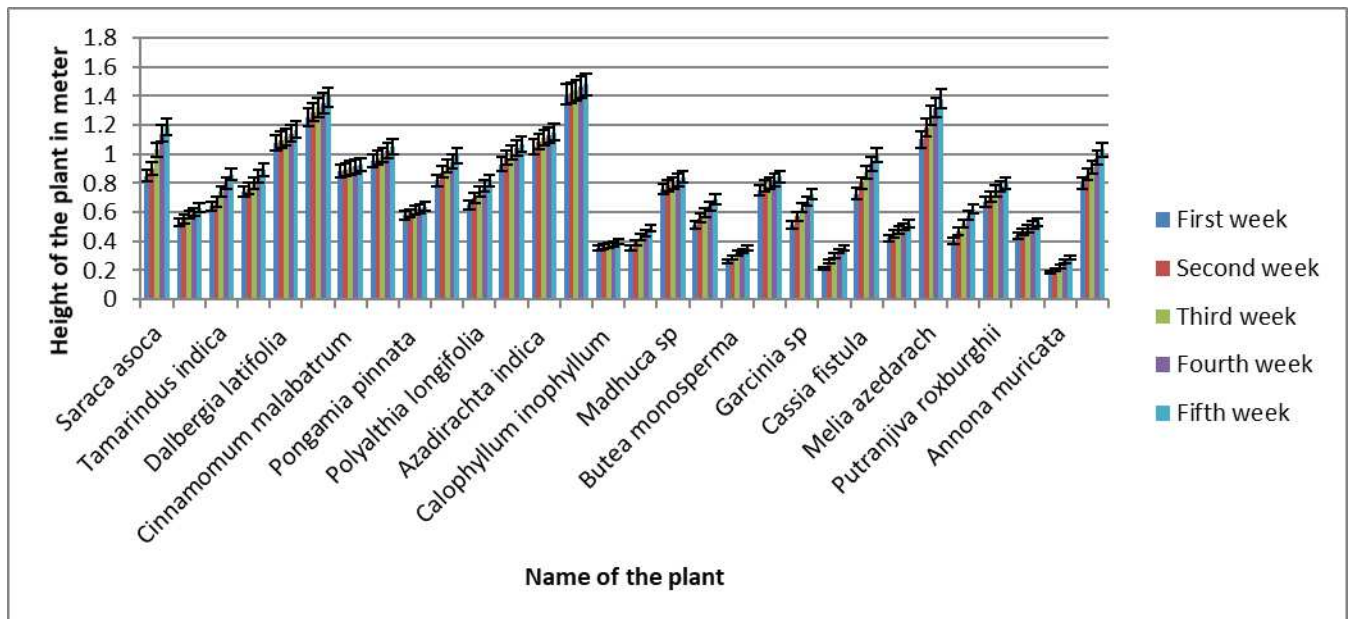


Figure 29. Height of the Trees in Site 3 (Dept. of Botany, Kariavattom)

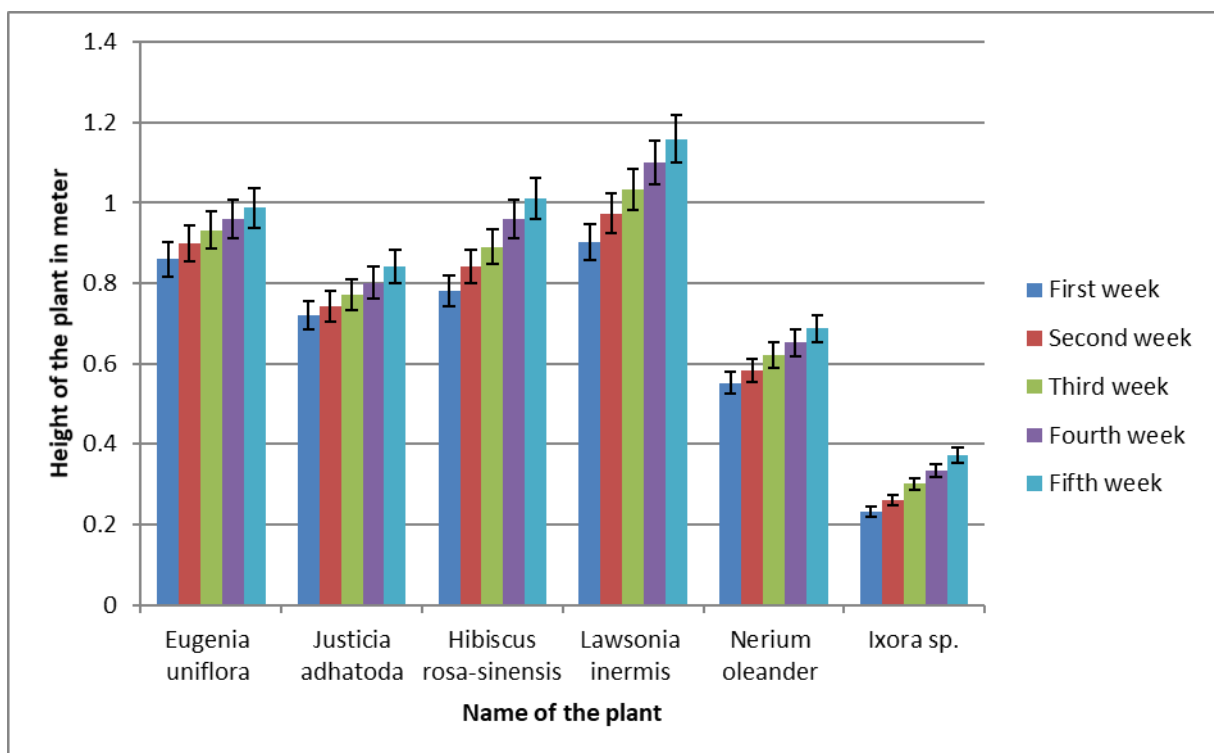


Figure 30. Height of the Shrubs in Site 3 (Dept. of Botany, Kariavattom)

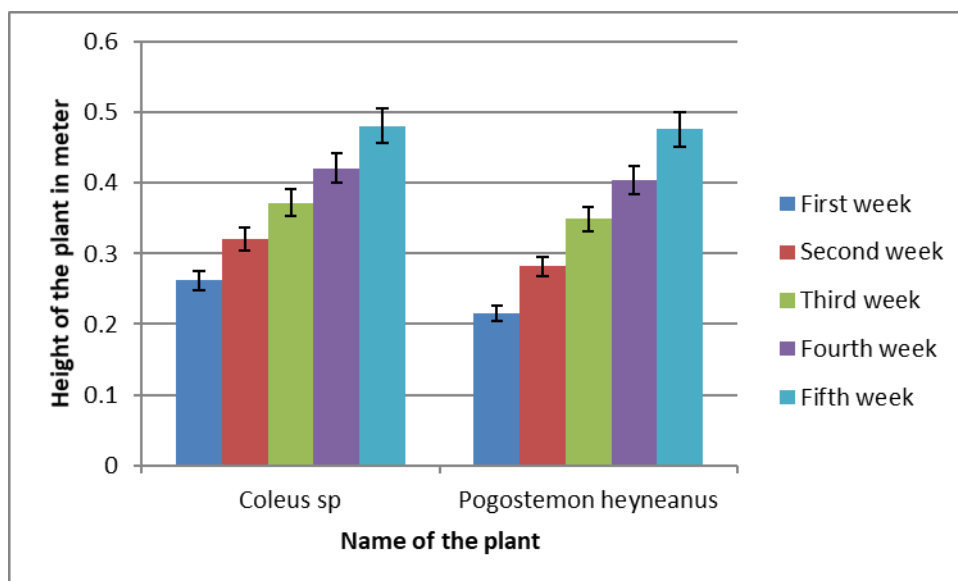


Figure 31. Height of the Herbs in Site 3 (Dept. of Botany, Kariavattom)

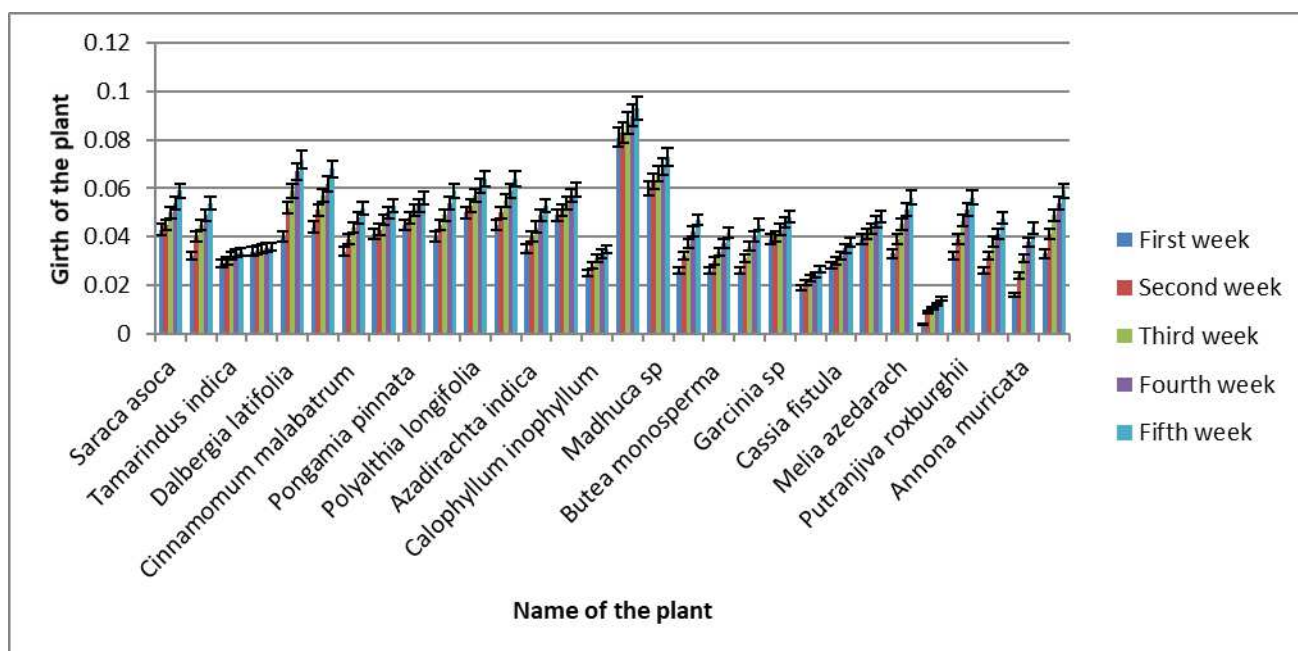


Figure 32. Girth of the Trees in Site 3 (Dept. of Botany, Kariavattom)

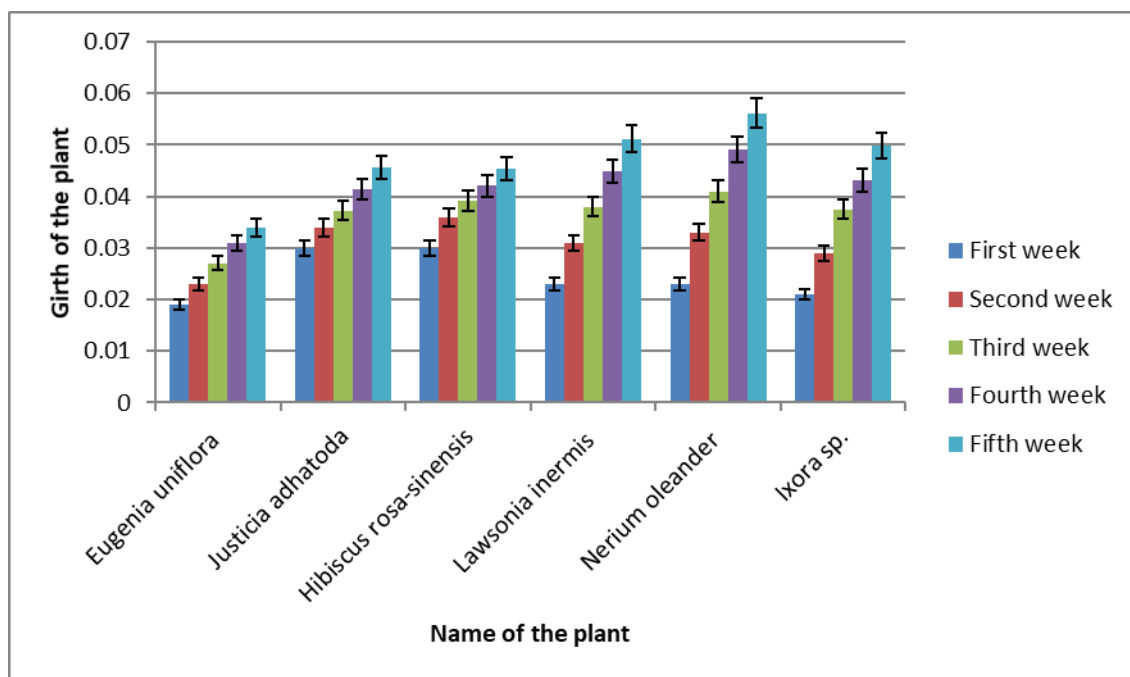


Figure 32. Girth of Shrubs in Site 3 (Dept. of Botany, Kariavattom)

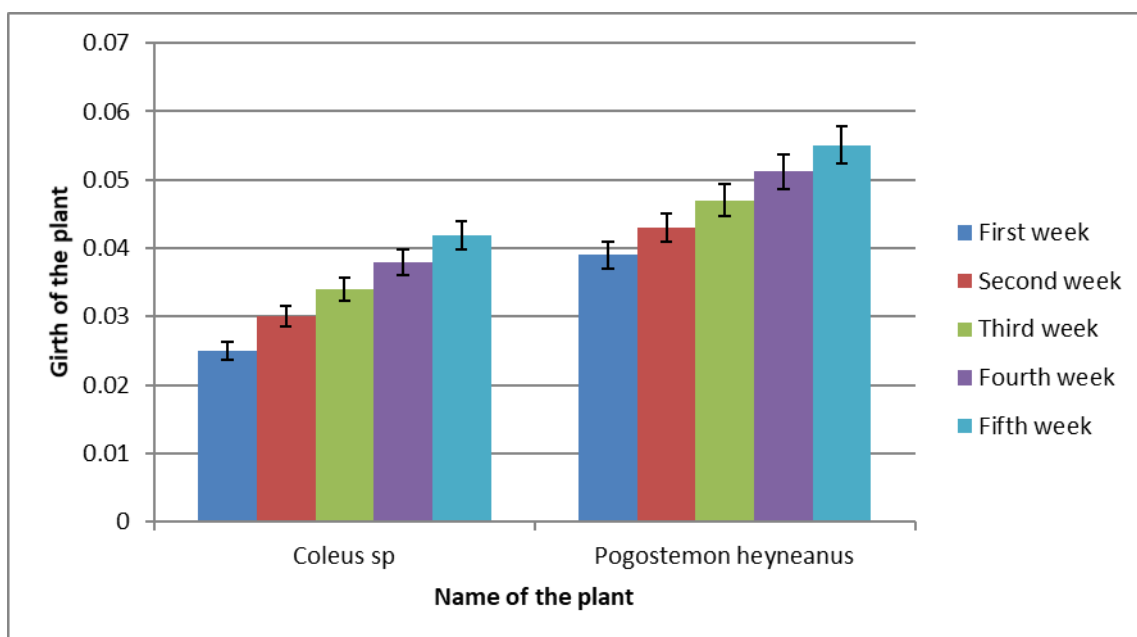


Figure 33. Girth of Shrubs in Site 3 (Dept. of Botany, Kariavattom)

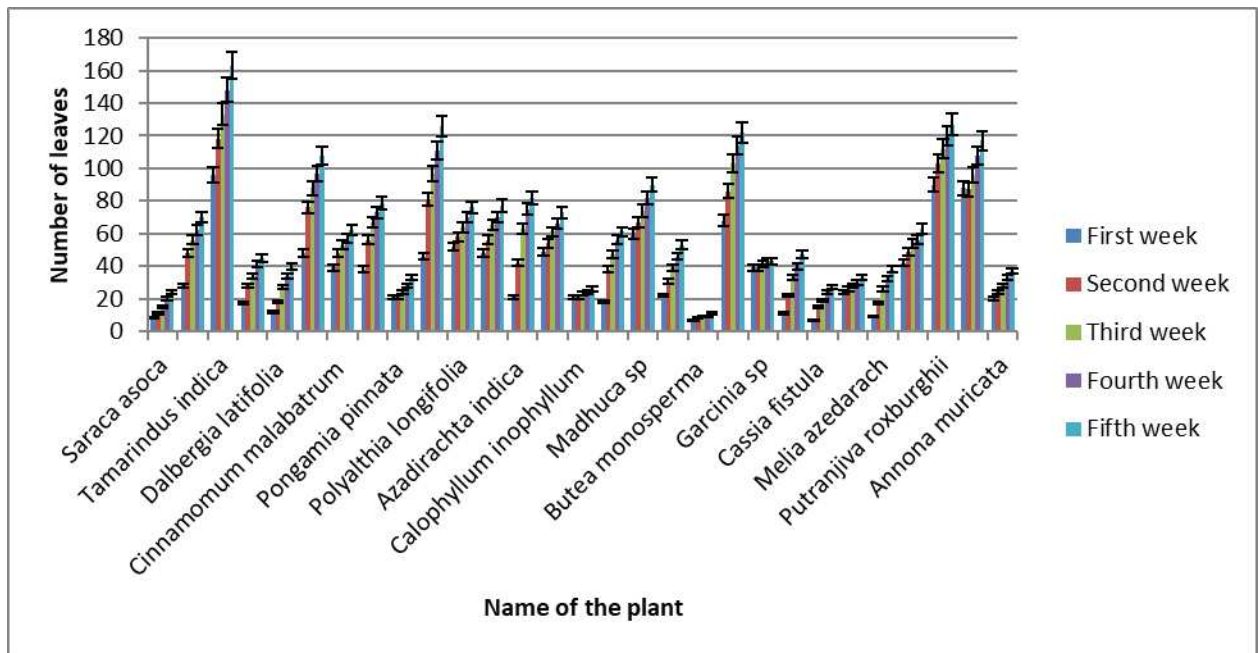


Figure 34. Number of leaves in Trees Site 3 (Dept. of Botany, Kariavattom)

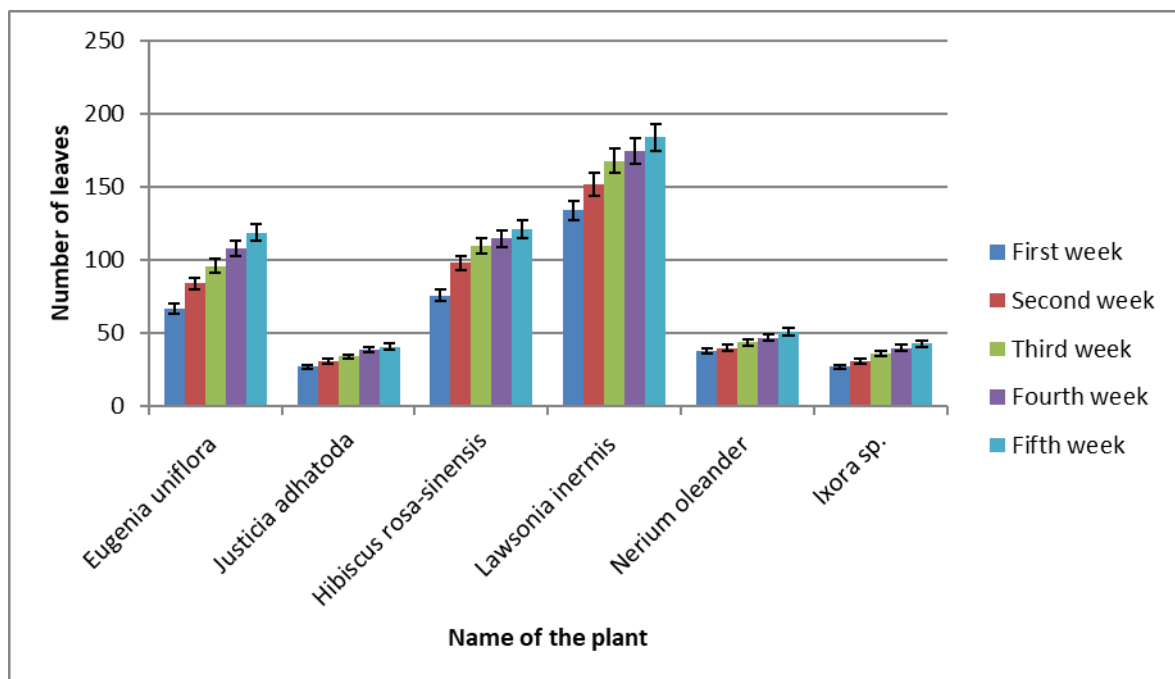


Figure 35. Number of leaves in Shrubs in Site 3 (Dept. of Botany, Kariavattom)

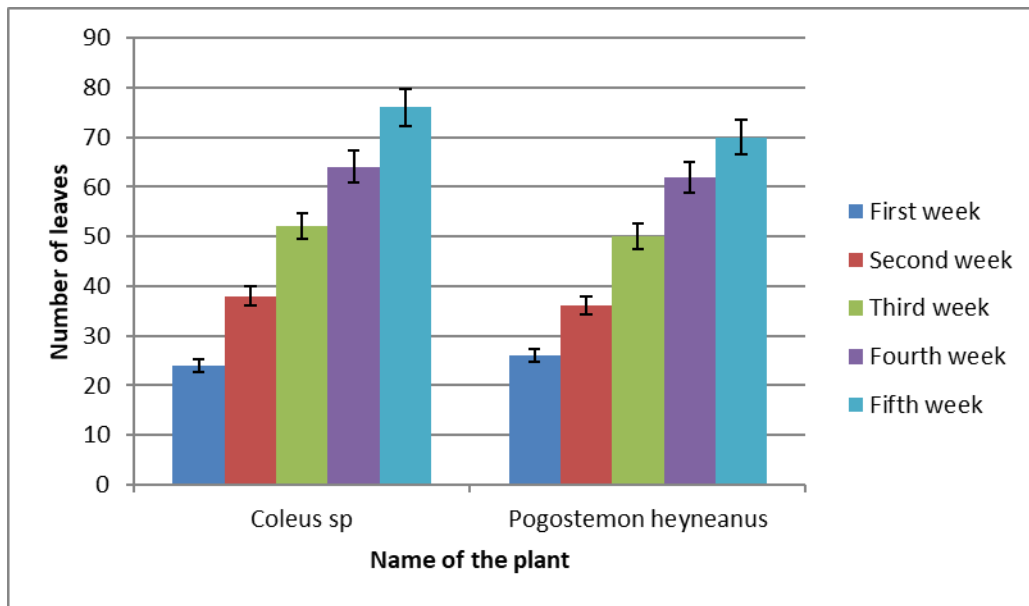


Figure 36. Number of leaves in Herbs in Site 3 (Dept. of Botany, Kariavattom)

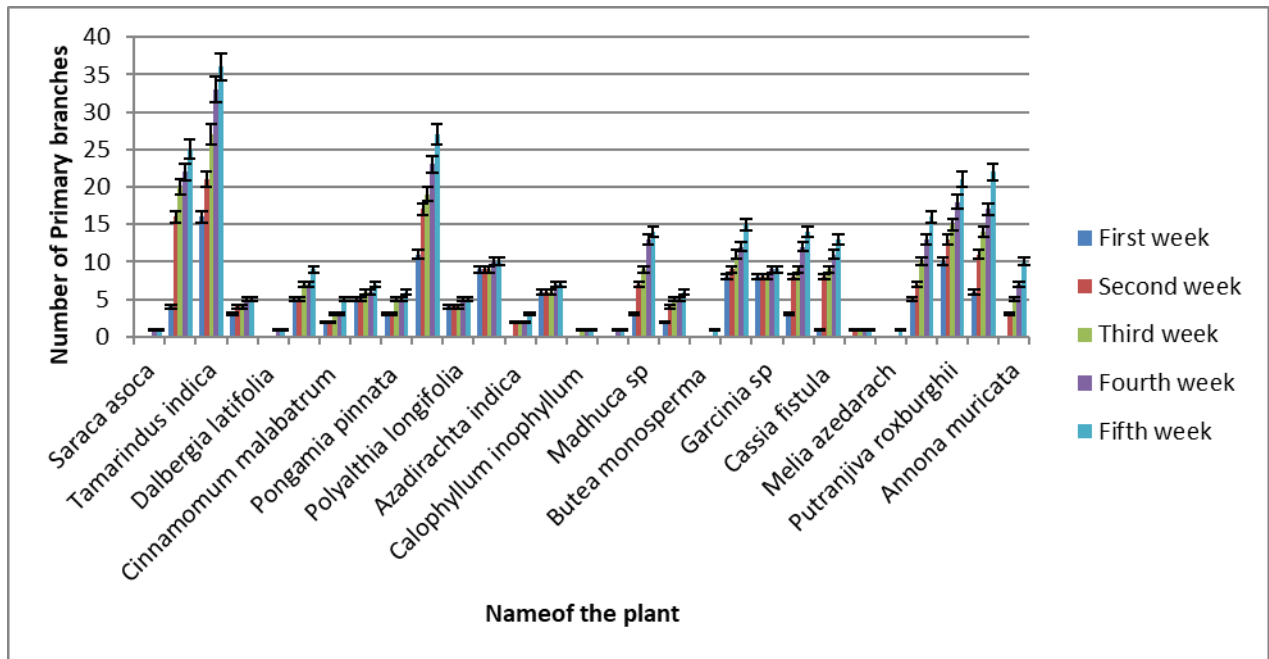


Figure 37. Number of Primary branches in Trees in Site 3 (Dept. of Botany, Kariavattom)

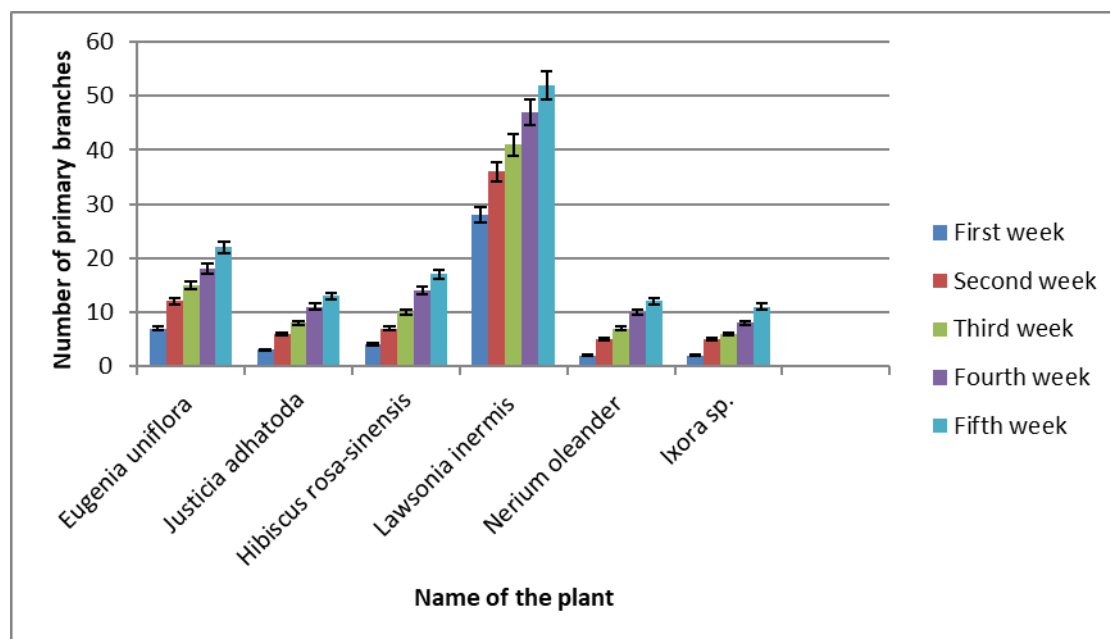


Figure 38. Number of Primary branches in Shrubs in Site 3 (Dept. of Botany, Kariavattom)

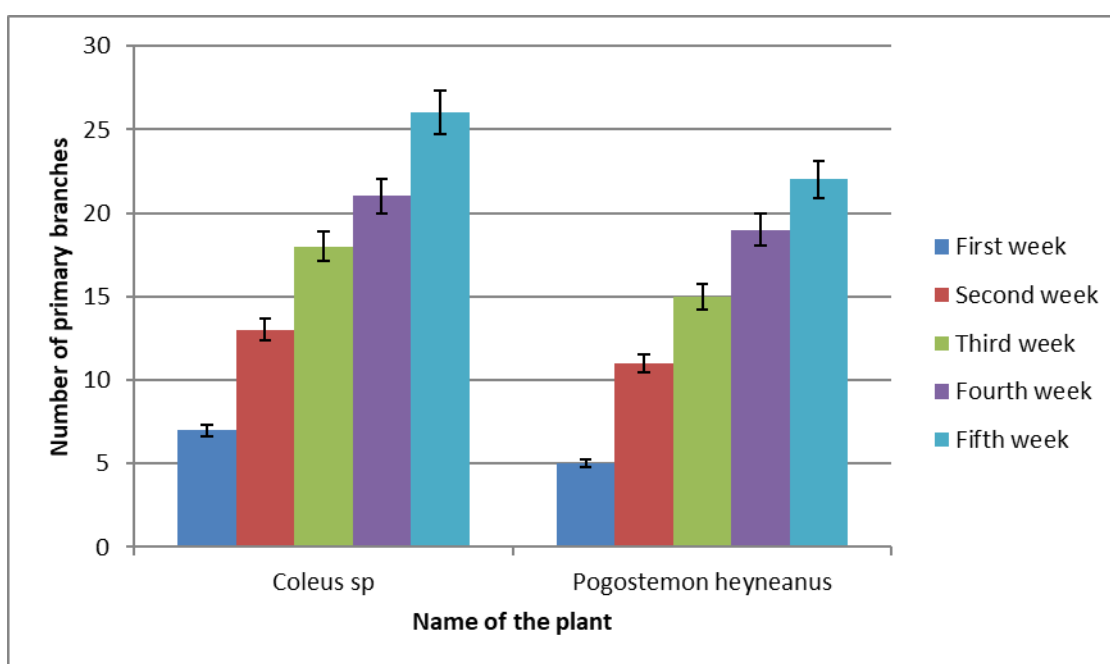


Figure 39. Number of Primary branches in Herbs in Site 3 (Dept. of Botany, Kariavattom)

Table 4. Growth parameters of the control plants after one year

Name of the plant	Height in meter (mean± SD)	Girth in meter (mean± SD)	Number of leaves (mean± SD)	Number of primary branches (mean± SD)
<i>Artocarpus heterophyllus</i>	1.9533±.07024	0.12467±0.004509	437.00±10.536	14.67±1.528
<i>Syzygium cumini</i>	1.2100±.07550	0.16333±0.004041	292.33±10.017	17.67±0.577
<i>Phyllanthus</i> sp.	2.2400±.06000	0.15000±0.004000	587.67±9.609	14.67±0.577
<i>Mangifera indica</i>	1.0167±.04509	0.14100±0.004000	217.00±5.000	12.67±0.577
<i>Morus alba</i>	3.6400±.02000	0.16600±0.004000	287.33±7.506	13.33±0.577
<i>Ixora coccinea</i>	0.7400±.02000	0.26333±0.015275	343.33±5.508	11.67±0.577

Table 5. List of Insects and plant visitors in three study sites

Sl. No.	Name of the Insects	Family	Interaction with plants
1	<i>Oxyopus</i> sp.	Oxyopidae	Pollinator/pest control
2	<i>Lycosa</i> sp.	Lycosidae	Pollinator/pest control
3	<i>Telomonia dimidiata</i>	Salticidae	Pollinator/pest control
4	<i>Agriope pulchella</i>	Araneidae	Pollinator/pest control
5	<i>Leptocorisa oratoria</i>	Alydidae	Pest/Pollinator
6	Short horned Grasshopper	Acrididae	Pest/Pollinator
7	<i>Ectobius sylvestris</i>	Ectobiidae	Pest/Pollinator
8	<i>Amata phegea</i>	Erebidae	Pest/Pollinator
9	<i>Micronia aculeata</i>	Uraniidae	Pollinator
10	<i>Ypthima huebneri</i>	Nymphalidae	Pollinator
11	<i>Rhyothemis variegata</i>	Libellulidae	Pollinator
12	Damselfly	Coenagrionidae	Pollinator
13	<i>Lampidus boeticus</i>	Lycaenidae	Pollinator
14	<i>Ichinemonid</i> wasp	Ichneumonidae	Pollinator
15	Cabbage butterfly	Pieridae	Pollinator
16	Red cotton bug	Pyrrhocoridae	Pest
17	<i>Graphium sarpedon</i>	Papilionidae	Pollinator
18	<i>Papilio demoleus</i>	Papilionidae	Pollinator
19	<i>Papilio polytes</i>	Papilionidae	Pollinator

Table 6. List of Fungi in three study sites

Sl. No.	Name of the fungus	Family
1	<i>Chlorophyllum molybdites</i>	Agaricaceae
2	<i>Leucocoprinus birnbaumii</i>	Agaricaceae
3	<i>Leucocoprinus bresadolae</i>	Agaricaceae
4	<i>Lenzites</i> sp.	Polyporaceae
5	<i>Pholiota</i> sp.	Strophariaceae
6	<i>Coprinellus</i> sp.	Psathyrellaceae
7	<i>Microporus xanthopus</i>	Polyporaceae
8	<i>Clathrus</i> sp.	Phallaceae
10	<i>Lepiota</i> sp.	Agaricaceae

PLATE 2: Insects and Pollinators in three study sites

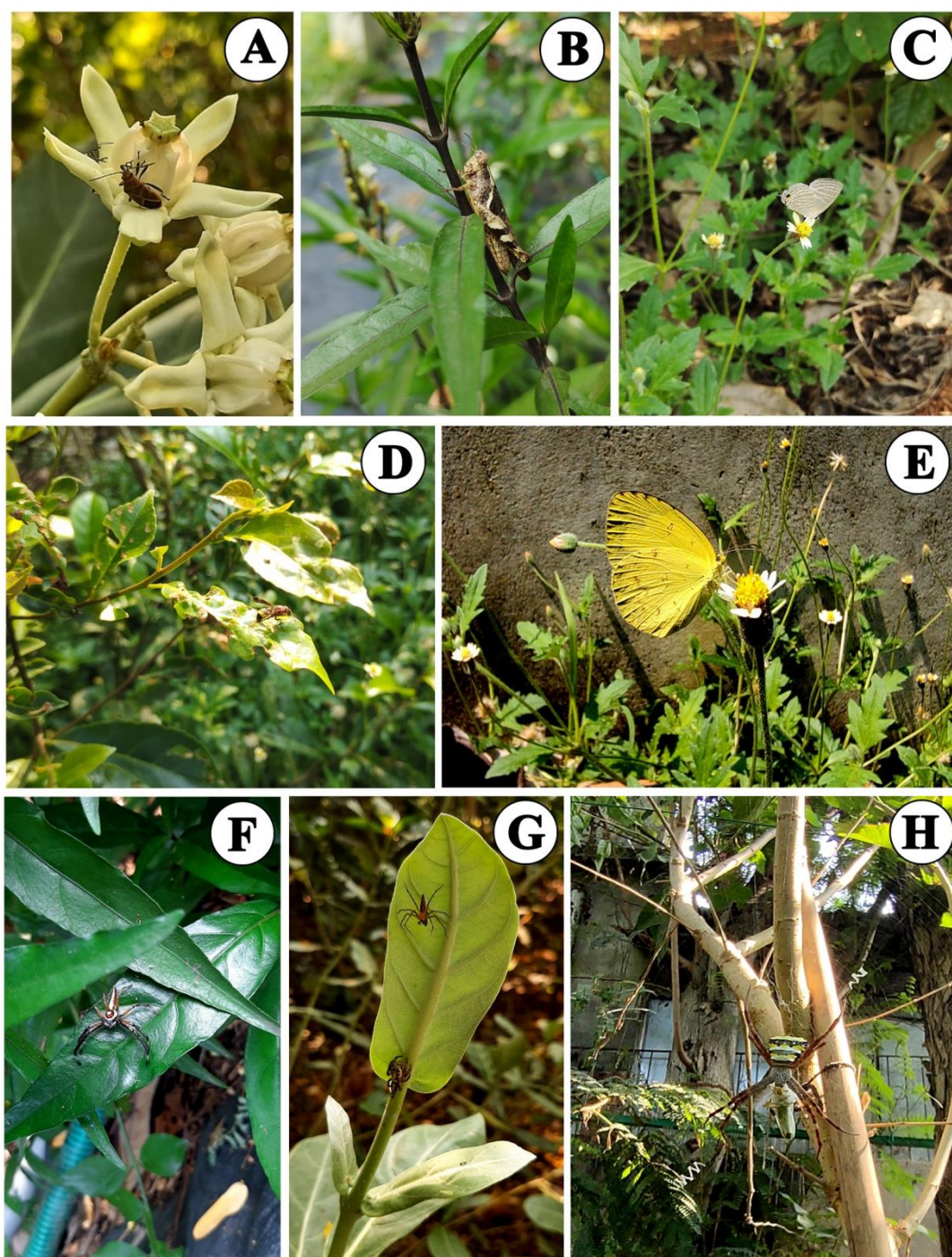
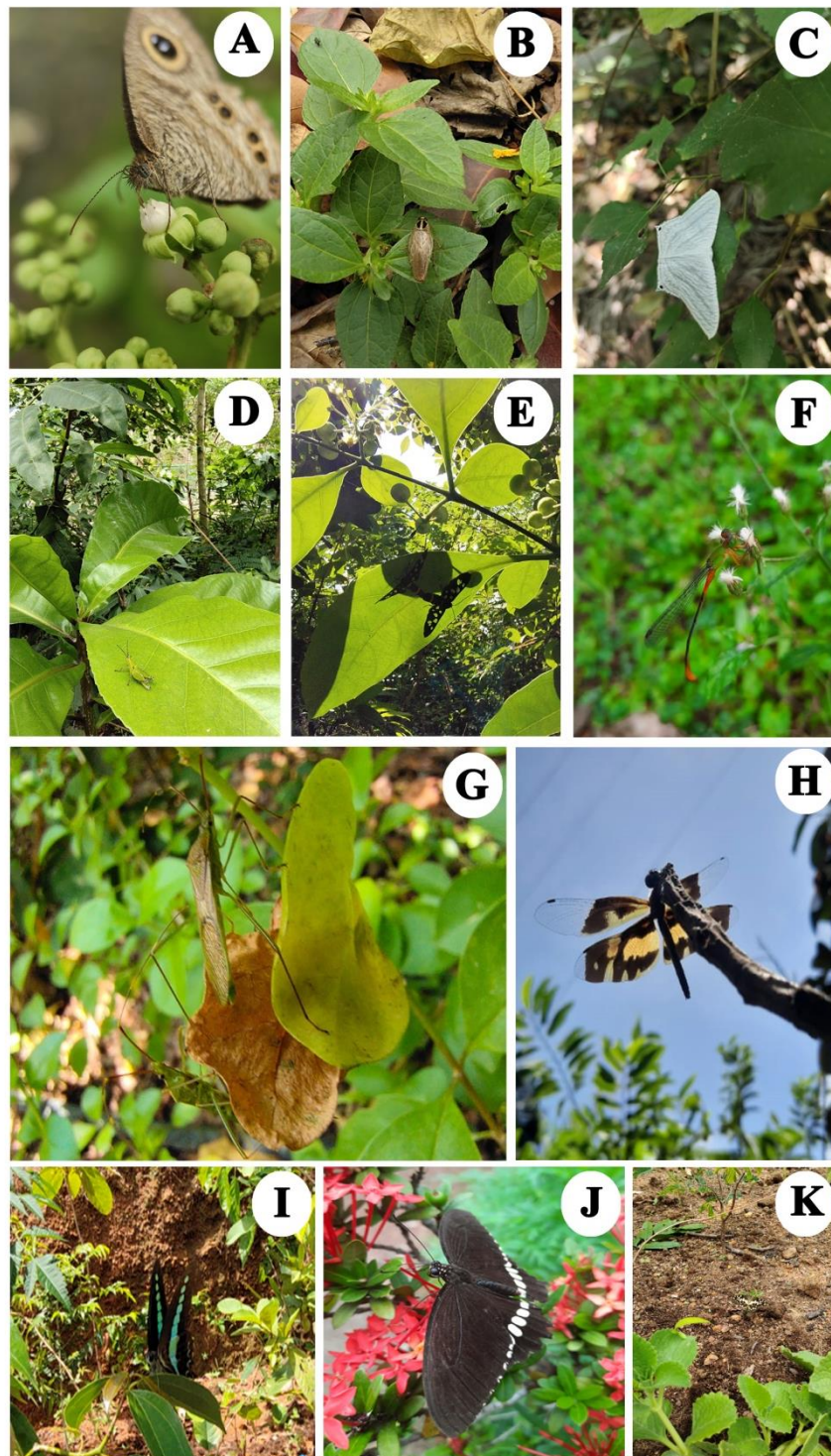


PLATE 3: Insects and Pollinators in three study sites



A: *Ypthima huebneri*, **B:** *Ectobius sylvestris*, **C:** *Micronia aculeata*
D: Short horned grasshopper, **E:** *Amata phegea*, **F:** Damselfly
G: *Leptocorisa oratoria*, **H:** *Rhyothemis variegata*, **I:** *Graphium sarpedon*,
J: *Papilio polytes*, **K:** *Papilio demoleus*

PLATE 4: Mushrooms in three study Sites



A:*Chlorophyllum molybdites*; **B:***Leucocoprinus birnbaumii*; **C:***Leucocoprinus bresadolae*; **D:***Lenzites* sp. **E:***Pholiota* sp.; **F:***Coprinellus* sp.; **G:***Lepiota* sp.; **H:***Microporus xanthopus*; **I:***Clathrus* sp

Conclusion

Many countries have recognized the value of rehabilitating degraded tropical areas, utilizing natural resources for sustainable development, and maintaining biodiversity. The Miyawaki method has been used successfully to restore the forest in many places. The advantage of this method is selecting as many native canopy species as possible based on the potential natural vegetation at each site by the phytosociological method. The Miyawaki micro-afforestation project aims to grow many trees in a limited space in a brief period of time and grow a forest similar to a natural forest. Replication of the model across Kerala, which has suffered floods, landslides and soil erosion, will help to restore the biodiversity in the state. The current study documents the vegetative and physiological maturity of selected plants in three Miyawaki sites, Kanakakunnu, Government High School Chala, and the Department of Botany Kariavattom. The result concluded that the plants grown in the Miyawaki field showed a faster growth rate. This micro afforestation method offers to address all the issues of soil erosion, which is a global problem. Urban areas and tourist areas with high environmental issues include climate change and air pollutions. Micro forest in urban areas is an effective and novel method to mitigate climate change. It also concluded that this micro-forest lower temperature and support local biodiversity included micro and macro flora and fauna. The present study proposes the positive impact of micro afforestation on the successful restoration of forest biodiversity and conservation of native plant species in urban areas. The results of the study could offer a chance to introduce the advantages of a new afforestation method, which helps reduce the time for a complete environmental restoration in a short period. Moreover, it also offers a model for conserving native plant species within their natural habitat, even in urban areas.

Acknowledgements

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